HPS Gaussian-Process Resonance Search

Selected Sections: Data, Statistical Method, Validation, and Results

Emrys Peets
Stanford University and SLAC National Accelerator Laboratory

Matthew Gignac
SLAC National Accelerator Laboratory

Eden Hsu
Stanford University and SLAC National Accelerator Laboratory

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Submitted by Emrys Peets. This excerpt comes from a collaborative Heavy Photon Search analysis note; all three note authors are listed above. HPS searches for a dark photon, A', as a narrow peak in the invariant mass of an electron–positron pair above a large, smoothly varying trident background. The selected sections follow that search from the detector configurations and e^+e^- spectra to a Gaussian-process prediction of the background beneath each blinded mass window, and finally to pseudoexperiments that test whether the procedure creates or absorbs a narrow signal. The numerical results and their statistical qualifications are unchanged from the analysis note.

Contents

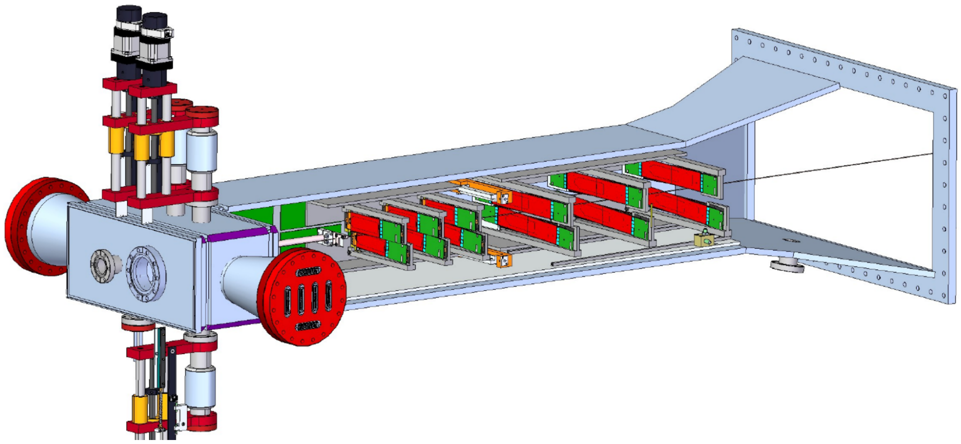
2 HPS apparatus and data sets

2.1 Detector configurations for the prompt search

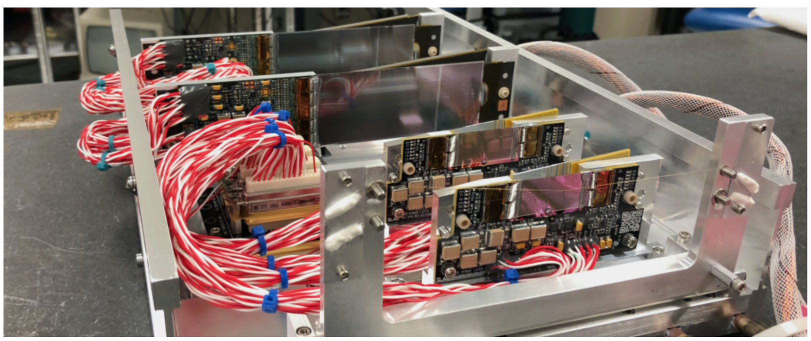
HPS is a forward spectrometer optimized for narrow e^+e^- resonances and displaced vertices in the high-rate environment immediately downstream of a thin tungsten target. Its prompt $A' \to e^+e^-$ sensitivity is set primarily by three subsystems: the Silicon Vertex Tracker (SVT), which determines the mass and vertex resolution; the electromagnetic calorimeter (ECal), which provides the trigger and track–cluster matching; and the beamline/target geometry, which defines the forward acceptance and the occupancy environment [1, 2].

The 2015 and 2016 engineering runs used the baseline six-station SVT. In that configuration, six axial/stereo tracking stations were distributed between $z = 10\,\mathrm{cm}$ to $90\,\mathrm{cm}$ downstream of the target, with the first active silicon only about $1.5\,\mathrm{mm}$ from the beam plane [3, 1]. The ECal geometry used in the engineering runs was already close to the final HPS layout: two $PbWO_4$ crystal halves with the highest-rate crystals nearest the beam removed, leaving the beam to pass between the two halves [3, 1].

Before the first physics run in 2019, HPS upgraded the front tracker. The 2019 and 2021 configurations added a seventh double layer at $z = 5\,\mathrm{cm}$ and used thin, slim-edge sensors with short split striplets [1, 4]. Moving the first precision measurements closer to the beam reduced material and occupancy while improving acceptance for low-mass, small-opening-angle pairs. The ECal crystal layout was unchanged, but a positron hodoscope was installed before the 2019 run to support a single-arm trigger for events near the ECal beam gap [1]. Figure 1 compares the tracker layouts used in the engineering and physics runs.



(a) Baseline six-layer SVT used in the 2015 and 2016 engineering runs [3].



(b) Front-layer upgrade used in the 2019 and 2021 physics runs, including the added seventh double layer and slim-edge front sensors [1, 4].

Figure 1: SVT configurations used in the prompt-resonance analyses. The 2015/2016 baseline tracker and the 2019/2021 tracker differ in both layer count and the design of the innermost stations. The ECal crystal layout was unchanged; the physics-run configuration added a positron hodoscope.

2.2 Data-taking campaigns and analysis scope

We consider two related analyses. The three-campaign combination uses the full 2015 and 2016 engineering-run samples together with a distributed 10% sample from the 2021 physics run. A later 2021-only study uses the same 2021 events but changes the lower edge of the GP fit support. The engineering data share the baseline tracker used in the published prompt searches, whereas the 2021 data were recorded with the upgraded tracker and positron hodoscope. Although the 2019 run used the same upgraded detector, no 2019 data enter either analysis.

Table 1 summarizes the run periods, analyzed samples, search intervals, and GP fit supports for the three-campaign combination. A mass hypothesis is tested only inside the search interval, whereas the GP is trained on the wider support after removing the local $2.25\,\sigma_m$ exclusion. The 2016 support begins at 30 MeV: with the five-bin rebinning used here, a nominal 35 MeV edge would leave no training-bin center below the exclusion at the 39 MeV search endpoint.

Run period	Beam energy	Sample	Search interval	GP fit support
2015 engineering run	1.056 GeV	$1170\,\mathrm{nb}^{-1}$	19 MeV to 90 MeV	14 MeV to 135 MeV
2016 engineering run	2.3 GeV	$10\,608\,\mathrm{nb}^{-1}$	39 MeV to 180 MeV	30 MeV to 210 MeV
2021 physics run	3.74 GeV	Native 10% sample from a $\sim 160\,\mathrm{pb}^{-1}$ parent run	50 MeV to 250 MeV	40 MeV to 300 MeV

Table 1: Data sets and mass domains in the three-campaign combination. The 2015 and 2016 inputs are the full engineering-run samples; the 2021 input is the distributed native 10% sample. The wider GP supports provide training sidebands beyond both ends of each search interval. The corresponding published prompt searches covered 19 MeV to 81 MeV in 2015 and 39 MeV to 179 MeV in 2016 [5, 2].

The 2021-only study changes only the lower support edge: it uses 36–300 MeV while retaining the 50–250 MeV search interval. The three-campaign combination in Table 1 continues to use 40–300 MeV for 2021.

Integrated luminosity and analyzed fraction do not enter the GP interpolation. They enter later, with the target, acceptance, efficiency, and radiative fraction, when an extracted prompt yield is converted to an equivalent ϵ^2 limit.

The 2015 and 2016 samples are fully unblinded. The 2021 input is distributed across the run rather than drawn from a contiguous block, so it samples varied run conditions while leaving most of the exposure unopened. The ROOT file does not contain enough exposure metadata to infer an integrated luminosity; “10%” describes how the sample was constructed. The three-campaign result combines the full 2015 and 2016 samples with this 2021 sample and is not the final result for the 2021 run.

Figure 2 compares the unit-normalized invariant-mass shapes. Figure 3 shows the same spectra in events per MeV and marks the fit supports and search intervals used in the three-campaign combination.

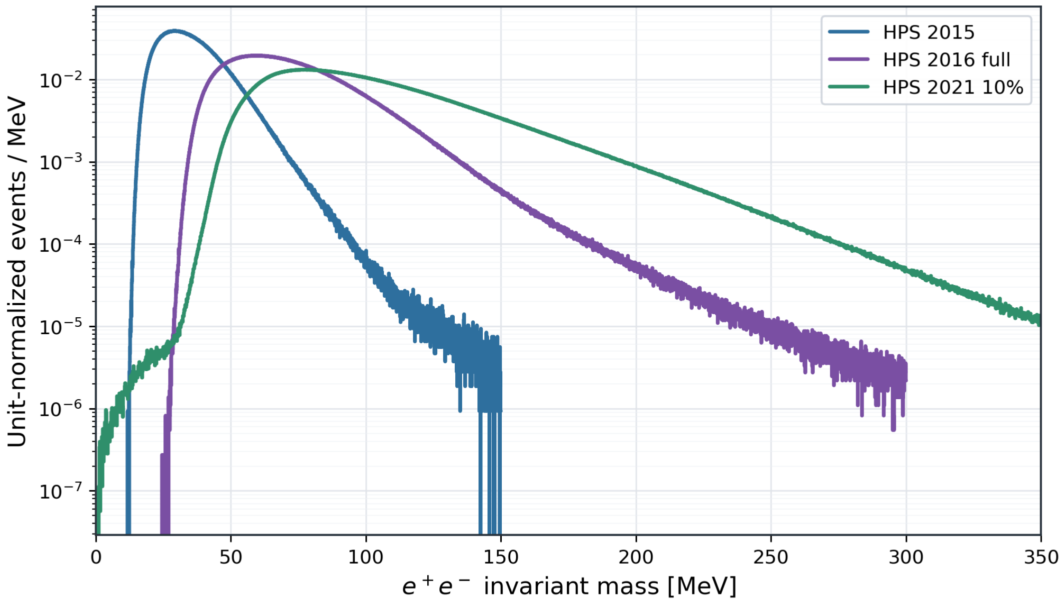


Figure 2: Unit-normalized invariant-mass shapes for the full 2015 and 2016 samples and the native 2021 10% sample. Each histogram is divided by its event count and bin width, so the comparison shows spectral shape rather than sample size.

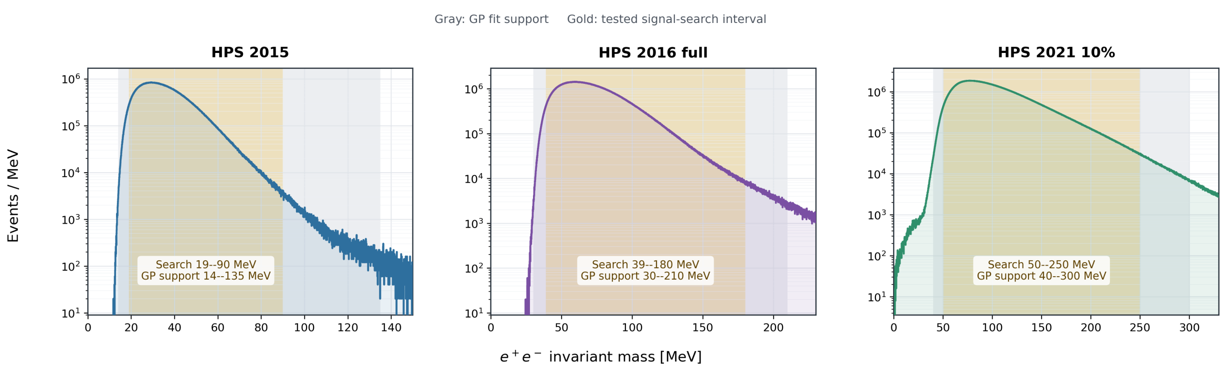


Figure 3: Invariant-mass distributions for the three-campaign combination. Gray shading marks the GP fit support and gold shading marks the narrower interval in which signal hypotheses are tested. The mass-dependent $2.25\,\sigma_m$ exclusion is removed from the GP support separately at every tested mass.

Although the three campaigns used different detector configurations and exposures, they enter the search through the same observable: a selected prompt e^+e^- invariant-mass spectrum. The next section explains the selections and the two inputs that connect a fitted resonance yield to the detector—mass resolution and radiative fraction.

3 Event selection, mass resolution, and radiative fraction

Each HPS input is an invariant-mass spectrum formed after a run-specific event selection. That selection determines the composition of the prompt e^+e^- sample; the detector response sets the width of a possible resonance, and the radiative fraction sets its normalization. Together, these ingredients connect the recorded events to the signal model used in the search.

3.1 Prompt event selection

Across the three HPS campaigns, the prompt selection follows the same experimental logic: oppositely charged tracks matched to ECal clusters, tight cluster-time coincidence, track-quality and vertex-reconstruction requirements, and momentum cuts that retain prompt radiative tridents while suppressing accidentals and converted wide-angle bremsstrahlung. The 2015 and 2016 selections are inherited from the published prompt searches. For 2021 we use the prompt target-constrained (prompt-TC) sample summarized in Table 2. It requires $p_{\text{sum}} > 2.8\,\text{GeV}$ and applies no additional vertex-fit χ^2 requirement. The stored hit count is three-dimensional and does not document a separate eight-hit two-dimensional electron requirement, so we do not include one here.

In Table 2, V_0 denotes a reconstructed e^+e^- vertex, GBL is the track-fit algorithm, FEE denotes the full-energy-electron veto, and TC means target constrained. Pair1, Single-2, and Single-3 are the trigger names used during the corresponding data-taking periods.

Table 2: Comparison of the prompt-resonance selections. The 2015 and 2016 columns follow the published and internal engineering-run summaries. For 2021, the preselection entries follow the full analysis note [6, Sec. 4.8 and App. A]; the prompt entries list the additional requirements documented for the native 10% prompt-TC sample.

Cut type	**2015 (Eng. run 1)**	**2016 (Eng. run 2)**	**2021 (native 10%)**
Trigger / topology	**Pre:** pairs-1 trigger; ECAL halves: $y^{\text{clu}}_{e^+} y^{\text{clu}}_{e^-} < 0$. **Prompt:** same.	**Pre:** Pair1 trigger bit; opposite ECal halves; the trigger does not encode track charge. **Prompt:** same.	**Pre:** Single-2 or Single-3 positron single-cluster trigger; Single-3 also requires the matched hodoscope condition used by the 2021 trigger menu. **Prompt:** no additional prompt-TC requirement.
Track momenta	**Pre:** $p(e^-) < 0.75\,E_{\text{beam}}$. **Prompt:** $p_{\text{sum}} > 0.8\,E_{\text{beam}}$ and $p_{\text{sum}} < 1.2\,E_{\text{beam}}$.	**Pre:** $p(e^\pm) < 2.15$ GeV; $p_{\text{sum}} < 1.2\,E_{\text{beam}}$. **Prompt:** $p_{\text{sum}} > 1.90$ GeV and $p_{\text{sum}} < 2.4$ GeV.	**Pre:** $p_{\text{trk}} > 0.4$ GeV for each track; FEE veto $p(e^-) < 2.9$ GeV; vertex momentum $p_{\text{vtx}} < 4.0$ GeV. **Prompt:** the prompt-TC sample requires $p_{\text{sum}} > 2.8$ GeV.
Track fit / hits	**Pre:** loose track–cluster match: $\chi^2_{\text{trk}-\text{clu}} < 10$. **Prompt:** GBL track $\chi^2_{\text{trk}} < 40$.	**Pre:** $\chi^2_{\text{trk}} < 12$; PID goodness < 10. **Prompt:** no additional prompt requirement is documented.	**Pre:** $\chi^2_{\text{trk}}/\text{ndf} < 20$ and $N_{\text{2D hits}} \geq 10$ for the tracks. **Prompt:** no additional 2D-hit requirement is established; the available field counts 3D hits.

Cut type	**2015 (Eng. run 1)**	**2016 (Eng. run 2)**	**2021 (native 10%)**
Track–cluster timing	**Pre:** — **Prompt:** $\lvert t_{\text{clu}} - t_{\text{trk}} - 43\text{ ns}\rvert < 4.5$ ns.	**Pre:** $\lvert t_{\text{trk}} - t_{\text{clu}}\rvert < 6$ ns. **Prompt:** no additional value stated.	**Pre:** full-note data timing windows: $\Delta(t_{e^-,\text{trk}}, t_{e^+,\text{clu}}) < 6.0$ ns, $\Delta(t_{e^+,\text{trk}}, t_{e^+,\text{clu}}) < 5.1$ ns, and $\Delta(t_{e^-,\text{trk}}, t_{e^+,\text{trk}}) < 7.8$ ns. **Prompt:** no additional prompt-TC timing requirement.
Cluster–cluster coincidence	**Pre:** — **Prompt:** $\lvert\Delta t_{\text{clu}-\text{clu}}\rvert < 2$ ns.	**Pre:** $\lvert\Delta t_{\text{clu}-\text{clu}}\rvert < 2$ ns. **Prompt:** $\lvert\Delta t_{\text{clu}-\text{clu}}\rvert < 1.43$ ns.	**Pre:** no separate offline cluster–cluster coincidence appears in the full preselection table; the Pair-0 hardware trigger has a cluster-pair timing window $\lvert\Delta t\rvert < 12$ ns and VTP prescale 100. **Prompt:** no additional prompt-TC offline coincidence requirement.
Vertex / V_0 quality	**Pre:** if multiple V_0 remain, choose the best χ^2_{vtx}. **Prompt:** $\chi^2_{\text{vtx}} < 75$ and $p(V_0) < 1.2\,E_{\text{beam}}$.	**Pre:** target-constrained V_0 candidates. **Prompt:** no additional prompt χ^2_{vtx} requirement is documented.	**Pre:** full-note upgraded-detector values: $p_{\text{vtx}} < 4.0$ GeV and $\chi^2_{\text{vtx}} < 20$. **Prompt:** no additional prompt-TC vertex-fit χ^2 requirement.
Positron cluster energy	**Pre:** — **Prompt:** —	**Pre:** — **Prompt:** —	**Pre:** full-note sanity cut $E^{\text{clu}}_{e^+} > 0.2$ GeV. **Prompt:** no additional prompt-TC positron-cluster threshold.
cWAB suppression	**Pre:** — **Prompt:** e^+ must have L1 and L2 hits; $d_0(e^+) < 1.1$ mm; and $(p_T^- - p_T^+)/(p_T^- + p_T^+) < 0.47$.	**Pre:** — **Prompt:** not applied in the final 2016 prompt selection summary.	**Pre:** no explicit cWAB-suppression cut is listed in the full 2021 preselection table. **Prompt:** no additional prompt-TC cWAB requirement.
Candidate multiplicity	**Pre:** unique-V_0 policy: keep the best χ^2_{vtx}. **Prompt:** enforced by the unique-V_0 logic.	**Pre:** — **Prompt:** require a single surviving V_0 candidate.	**Pre:** exactly one selected good vertex: $N_{\text{vtx}} > 0$ and $N_{\text{vtx}} < 2$. **Prompt:** no additional prompt-TC multiplicity requirement.

The 2021 preselection entries follow the full analysis note [6, Sec. 4.8 and App. A]. The native 10% spectrum comes from the prompt-TC sample. The GPR fit adds no event-level cuts. Each campaign enters the analysis through its selected invariant-mass histogram and the detector-response and normalization inputs for the same event category.

The engineering-run notes use *preselection* for the broad reconstruction sample and *prompt* *selection* for the tighter resonance-search sample. The 2021 input is closer to the latter because the prompt-TC sample retains the high-p_{sum} side. Table 2 keeps that distinction without adding an unrecorded two-dimensional hit requirement.

3.2 2015 mass resolution

The 2015 signal width is calibrated with Møller data and propagated across mass with simulated A' samples [7]. The Møller peak fixes the detector resolution at a known mass, and the data–MC momentum-resolution difference determines the additional smearing applied to the simulation. Rescaling the fitted A' widths to that data anchor gives

$$\sigma_{2015}(m) = -9.22 \times 10^{-5} + 5.322 \times 10^{-2}\, m, \tag{1}$$

with m and σ_{2015} expressed in GeV.

The Møller measurement supplies the absolute response, while the simulated mass grid supplies its mass dependence. The associated uncertainty therefore reflects the Møller calibration, the target position, and the residual momentum-response mismodelling [5, 7]. At each test mass this curve sets both the Gaussian signal width and the width of the excluded training interval.

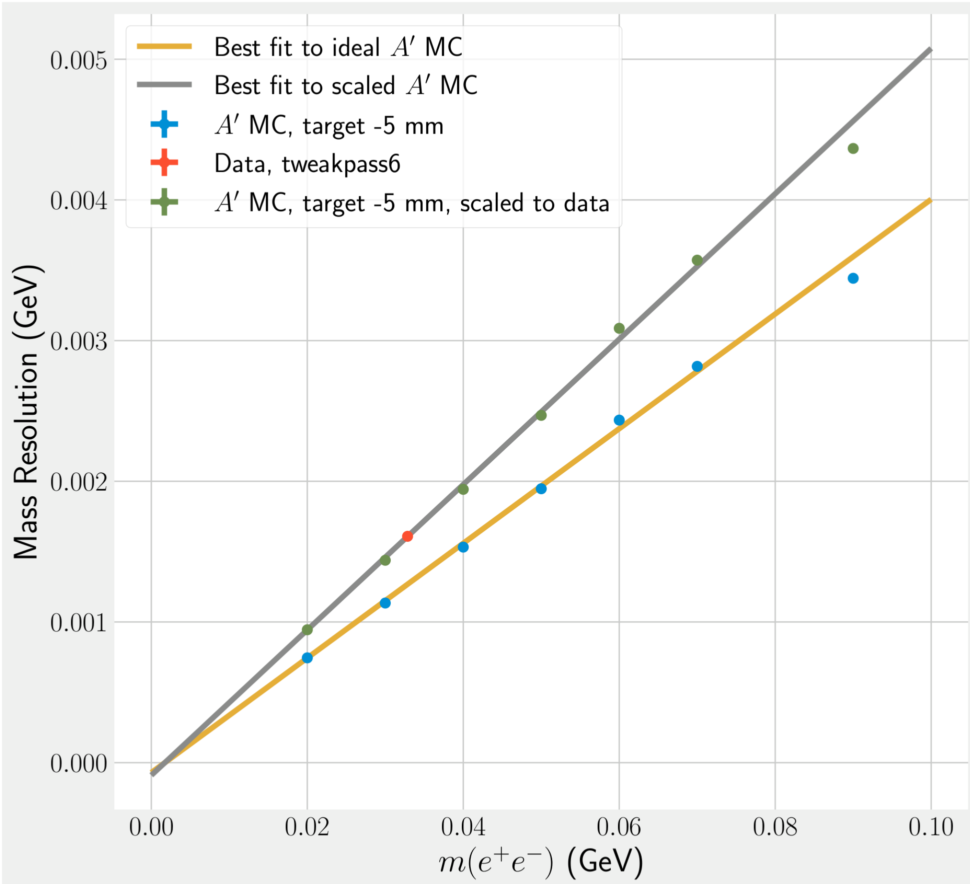


Figure 4: 2015 mass-resolution parameterization from scaled A' MC and Møller data [7]. The Møller point anchors the absolute detector response in data, while the scaled A' simulation supplies the mass dependence across the search window. Because the prompt dark-photon width is negligible, this curve directly sets the resonance width assumed in the search and therefore the local background entering the limit.

3.3 2016 mass resolution

The 2016 calibration follows the same strategy over a broader mass range [3, 2]. Møller events fix the absolute response at 48.5 MeV; full-energy-electron momentum peaks determine the extra smearing needed to match data; and smeared A' samples carry that calibration across the search range. A fourth-order polynomial fit to those samples defines the 2016 signal model,

$$\begin{aligned}\sigma_{2016}(m) = 3.8\times10^{-4} + 4.1\times10^{-2}m - 2.7\times10^{-1}m^2 \\ + 3.49\,m^3 - 11.11\,m^4, \qquad m \le 0.18\,\text{GeV},\end{aligned} \tag{2}$$

Earlier extended-range studies continued the polynomial linearly above $m_0 = 0.18\,\text{GeV}$ with slope 0.0239 to avoid its high-mass turnover. The present 2016 scan ends at 180 MeV, close to the published 39 MeV to 179 MeV interval, so that continuation is not used. The 2015 scan similarly remains near its published 19 MeV to 81 MeV interval, with only the upper endpoint extended to 90 MeV.

The smearing correction reconciles the Møller and simulated A' widths with the observed tracking response [3, 2]. Because the resolution worsens with mass, both the signal template and the excluded training window broaden, increasing the prompt background beneath a narrow signal hypothesis.

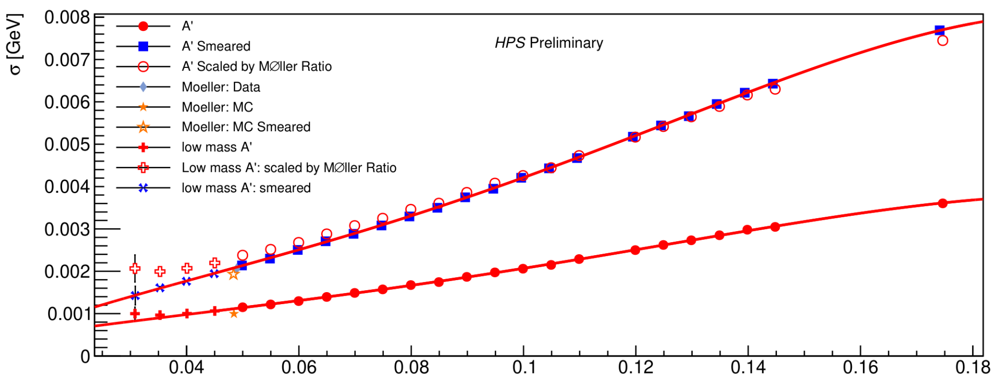


Figure 5: 2016 smeared-A' and Møller-based mass-resolution summary [3]. The smeared A' points show how the data-driven momentum-smearing correction is propagated away from the single Møller calibration point. The worsening resolution with mass broadens the signal window and is one of the main detector effects limiting prompt sensitivity at higher mass.

3.4 2021 mass resolution

The upgraded 2019/2021 tracker requires a separate resolution parameterization. We use the inclusive target-constrained V0 fit without hit requirements shown in Figure 6. Before the final data–MC broadening, the red curve is

$$\sigma_{2021}^{\mathrm{TC,raw}}(m_{\mathrm{MeV}}) = \left(1.4786 - 1.1 \times 10^{-3} m_{\mathrm{MeV}} + 6.87 \times 10^{-5} m_{\mathrm{MeV}}^2\right)\ \mathrm{MeV}, \tag{3}$$

where m_{MeV} is the numerical mass value in MeV. The analysis then applies the target-constrained data/MC scale derived from the 2021 smearing study,

$$s_{\mathrm{TC}} = \frac{2.186}{1.745} \simeq 1.25, \qquad \sigma_{2021}^{\mathrm{TC,smear}}(m) = s_{\mathrm{TC}}\, \sigma_{2021}^{\mathrm{TC,raw}}(m). \tag{4}$$

The scale comes from the target-constrained row after hit and curvature (Ω) smearing are applied to the simulation. Equation 4 then sets the 2021 signal width and the corresponding excluded training interval at each mass.

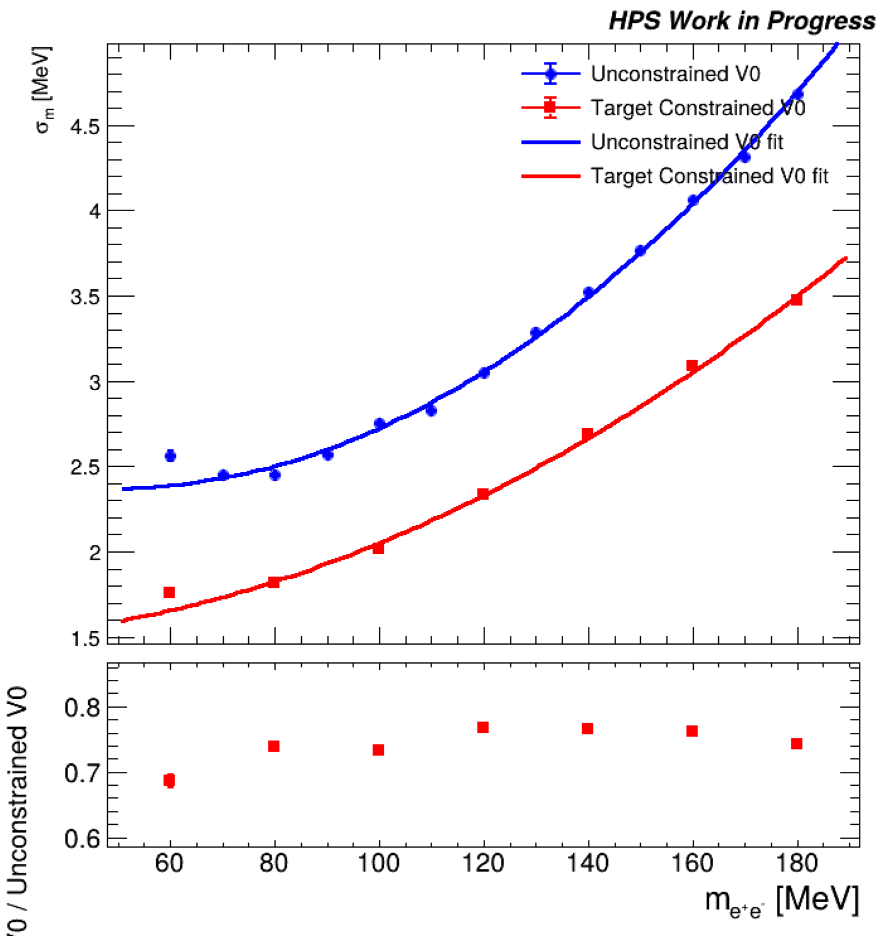


Figure 6: 2021 mass-resolution summary used for the prompt analysis. The red curve is the inclusive target-constrained V0 quadratic fit without hit requirements before the TC data/MC broadening in Eq. 4. The blue curve shows the corresponding unconstrained V0 reference fit included in the underlying study for comparison.

Vertex Type	Hit Smear Only (MeV)	Omega + Hit Smear (MeV)	Data (MeV)
UC	2.329 ± 0.013	2.463 ± 0.014	2.946 ± 0.036
BSC	2.092 ± 0.012	2.193 ± 0.012	2.569 ± 0.035
TC	1.590 ± 0.009	1.745 ± 0.010	2.186 ± 0.031

Figure 7: 2021 Møller run data/MC mass-resolution comparison used to set the smearing scale. Here UC denotes the unconstrained vertex sample, BSC denotes the beamspot-constrained vertex sample, and TC denotes the target-constrained vertex sample. The TC row is the most relevant row for the prompt analysis; comparing the data value 2.186 MeV to the Ω+hit-smeared Monte Carlo value 1.745 MeV gives the applied scale factor $2.186/1.745 \simeq 1.25$.

Figure 8 compares the three resolution curves used to construct the signal templates and excluded training windows over their respective scan ranges.

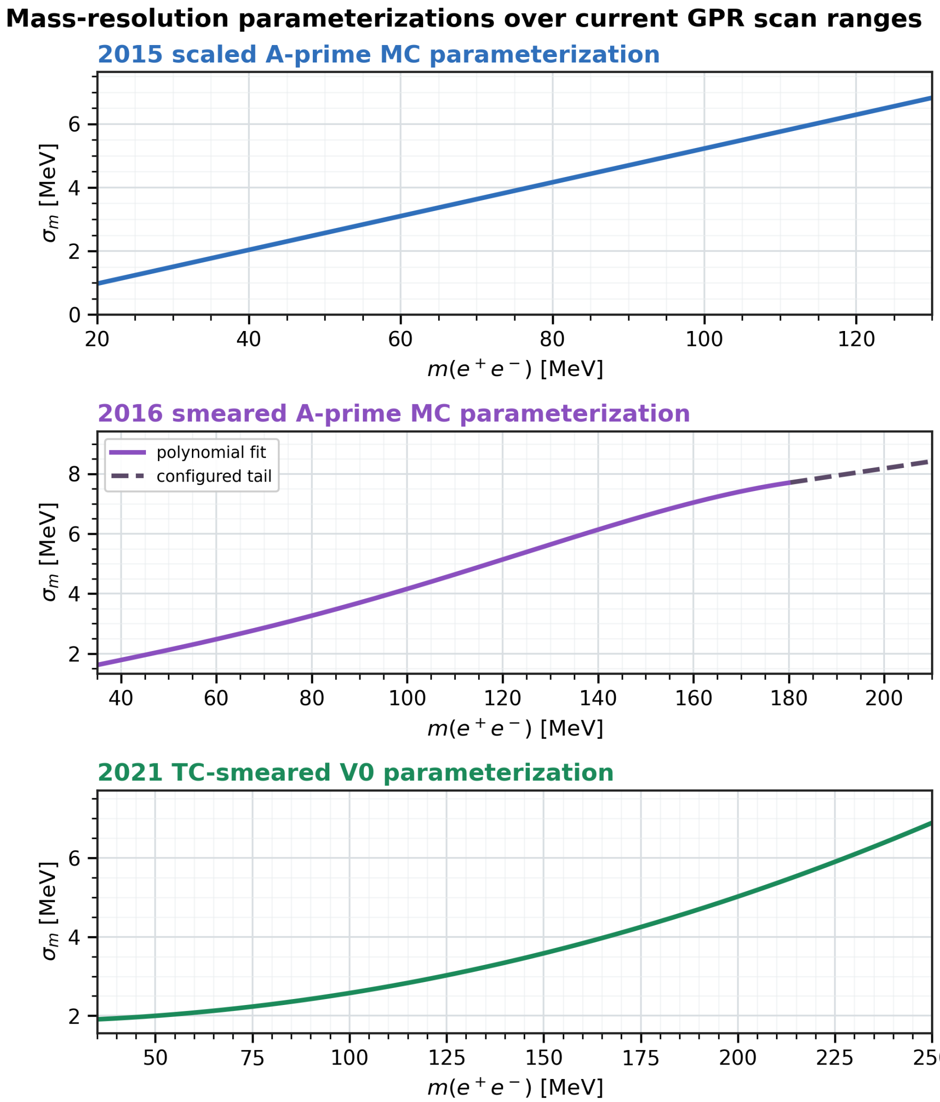


Figure 8: Mass-resolution parameterizations used by the analysis over the adopted scan ranges. The 2016 scan ends at the polynomial boundary of 180 MeV, so its configured high-mass continuation is not used. The 2021 panel shows the target-constrained V0 parameterization after multiplication by the TC smearing scale $s_{\mathrm{TC}} = 2.186/1.745 \simeq 1.25$ and extended to 250 MeV.

3.5 Radiative fraction and the ε^2 normalization

The radiative fraction encodes the fraction of the selected prompt background arising from the purely radiative trident process,

$$f_{\mathrm{rad},d}(m) = \frac{dN_{\gamma^*,d}/dm}{dN_{\mathrm{tri},d}/dm + dN_{\mathrm{wab},d}/dm}, \tag{5}$$

where the numerator is the selected radiative-trident component and the denominator is the full selected prompt background for dataset d [5, 2]. Because heavy-photon production is normalized to the radiative trident rate, f_{rad} is the physics quantity that converts a fitted prompt signal yield into a limit on the coupling. With this notation,

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi\, m\, f_{\mathrm{rad},d}(m)\, \rho_d(m)}{2\alpha_{\mathrm{EM}}}, \qquad A_d = K_d(m)\, \varepsilon^2, \tag{6}$$

where $\rho_d(m)$ is the selected prompt-spectrum density in dataset d. The engineering-run inputs follow the corresponding prompt searches, while the 2021 normalization remains provisional for the native 10% sample.

3.5.1 2015 radiative fraction

The 2015 prompt search used a constant approximation,

$$f_{\mathrm{rad},2015}(m) = 0.085, \tag{7}$$

which the analysis retains as its 2015 baseline. The dedicated 2015 radiative-fraction study compares the selected radiative-trident component with the total prompt background and adopts the red horizontal line at 8.5% as a conservative description across the search interval. The constant comes from the internal HPS normalization study rather than from the GP fit and is retained for consistency with the published 2015 analysis.

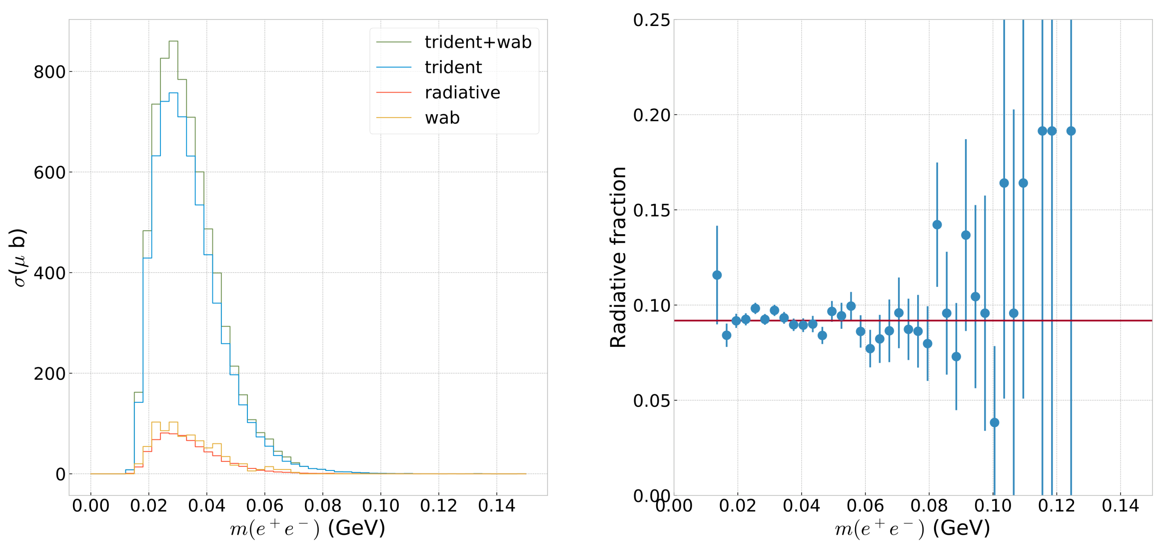


Figure 9: Dedicated 2015 radiative-fraction inputs. The left panel shows the selected cross-section composition, while the right panel shows the radiative fraction used for normalization. The red horizontal line denotes the 8.5% constant adopted as the conservative fit across the 2015 prompt-search dataset. This quantity converts the measured prompt spectrum into the expected A' normalization, so it affects the final ε^2 interpretation rather than the local spectral fit itself.

3.5.2 2016 radiative fraction

The 2016 prompt search used a fifth-order polynomial derived from radiative, trident, and converted-WAB MC after the full prompt selection. Those components determine the mass-dependent conversion from fitted signal yield to ε^2 over the 2016 search range. For the 2016 input, we instead use the constant approximation

$$f_{\mathrm{rad},2016}(m) = 0.05, \tag{8}$$

which is numerically stable over the wider scan range. Figure 10 shows the mass-dependent simulation used in the published determination; the constant used here is a working numerical approximation to that richer model.

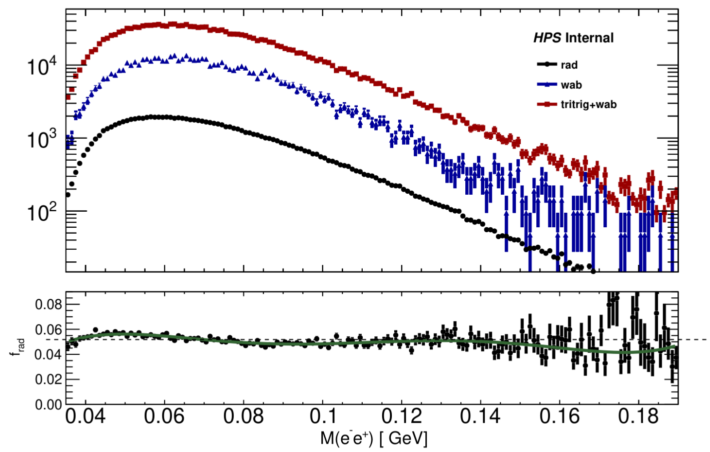


Figure 30: The differential rates in units of MeV^{-1} for all the MC samples which go into the radiative fraction. The rad component is a function of the true mass of the generator level gamma* for the event. The WAB and tritrig components are a function of the invariant mass of the selected V0 candidate. The bottom panel shows the radiative fraction, the green line is the fifth order polynomial fit, and the dotted line is the zeroth order polynomial fit.

Figure 10: 2016 radiative-fraction study from radiative, trident, and converted-WAB MC [3]. The simulation gives an intrinsically mass-dependent normalization, although this analysis uses the constant $f_{\mathrm{rad},2016} = 0.05$ for numerical stability.

3.5.3 2021 radiative fraction

For the native 2021 10% sample, we use the provisional value

$$f_{\mathrm{rad},2021}(m) = 0.05. \tag{9}$$

The 2021 full analysis note constructs the ratio from selected radiative-trident, tritrig, and WAB samples after preselection and a high-momentum requirement quoted as $p_{\mathrm{vtx}} > 2.8$ GeV. A fit over $0.05 < m < 0.25$ GeV gives the constant average $f_{\mathrm{rad}}^{\mathrm{const}} = 0.0496$ [6, Sec. 6.1, Figs. 92–93, Eqs. 25–26]. Thus 0.05 is a reasonable approximation over most of the 2021 search range. Before it is used for a final 2021 result, this normalization must be checked directly against the prompt-TC p_{sum} selection and its threshold convention; the mass-dependent parameterization provides a useful cross-check.

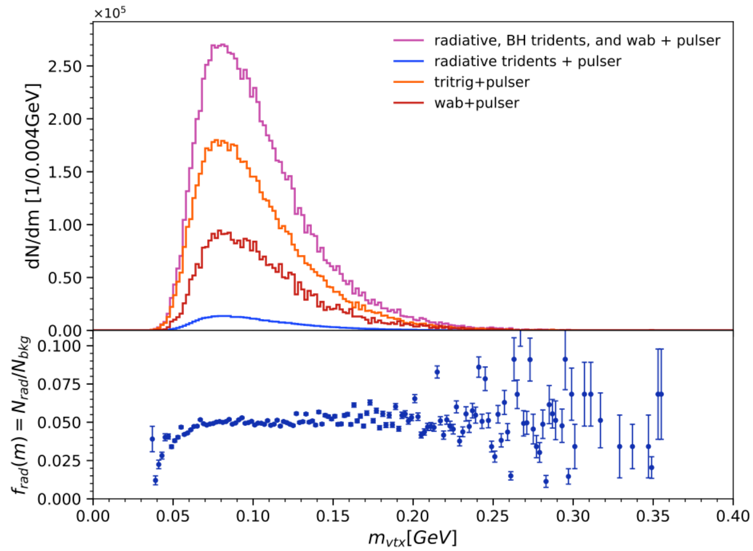


Figure 92: Invariant mass distributions of background samples. The lower panel shows the ratio of radiative tridents to the combined tritrig and WAB samples. This is referred to as the radiative fraction.

Figure 11: 2021 selected-mass distributions used to form the radiative fraction in the full-analysis note [6]. The upper panel shows the selected radiative, tritrig, and WAB components, while the lower panel gives $f_{\mathrm{rad}}(m) = N_{\mathrm{rad}}/N_{\mathrm{bkg}}$.

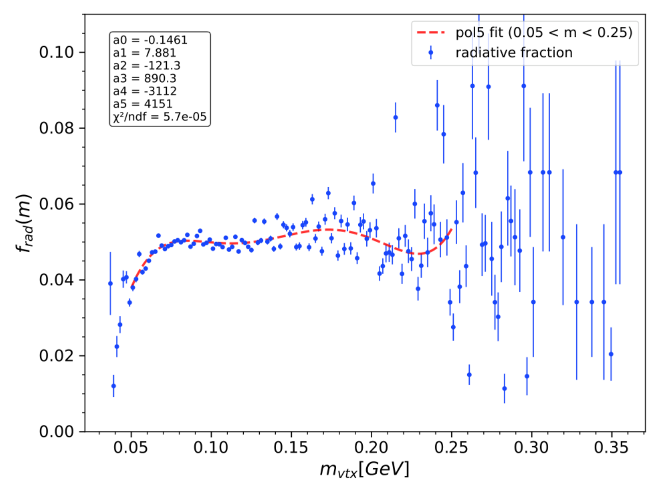


Figure 93: 5th-order polynomial fit to the radiative fraction. The fit range and results are indicated in the figure.

Figure 12: Fifth-order 2021 full-analysis-note fit to the binned radiative fraction [6]. The fitted curve is close to the constant value 0.05 across most of the range used by the 2021 analysis.

3.6 Systematic penalties from the radiative fraction

The analysis applies a data-set-dependent radiative-fraction penalty,

$$f^{\mathrm{eff}}_{\mathrm{rad},d}(m) = (1 - \delta_{f,d})\, f_{\mathrm{rad},d}(m). \tag{10}$$

The penalty enters only when signal yield is converted to ε^2. It changes the normalization of the inferred coupling without altering the blind-window likelihood.

The nominal configuration uses

$$(\delta_{f,2015}, \delta_{f,2016}, \delta_{f,2021}) = (0.070, 0.070, 0.046). \tag{11}$$

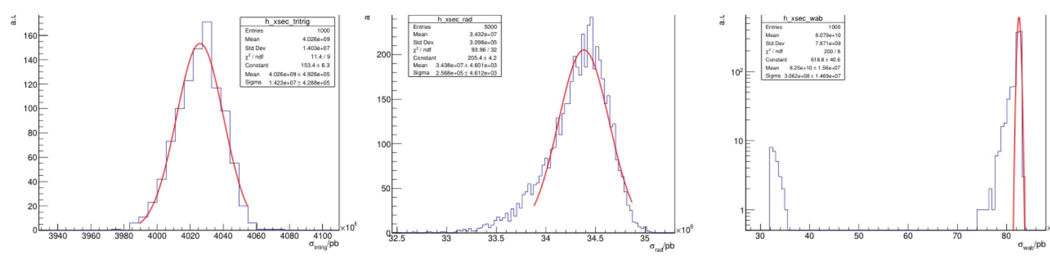
For a fixed signal-yield limit, these shifts weaken the coupling-squared limit by factors $1/0.93$ for 2015 and 2016 and $1/0.954$ for 2021. In the combined fit each $K_d(m)$ is reduced by its own amount, so a larger shared ε^2 is required to produce the same yields. The factors are fixed, one-sided normalization shifts; they do not enter the GP interpolation and are not profiled as nuisance parameters.

The 7% engineering-run penalties reflect the several-percent uncertainty in the 2015 radiative and total e^+e^- composition and the corresponding 2016 uncertainty from the selected radiative, trident, and converted-WAB inputs [7, 2]. We retain 7% as a conservative first-order treatment for both engineering runs. The 2021 term uses the upgraded-detector estimate below.

The 2021 full analysis note assigns 3.5% from the RAD and WAB cross sections shown in Fig. 13, using their quoted widths and the relative normalization $R = 0.244$ [6, Sec. 9.1, Fig. 161]. It assigns a further 3% for preselection and radiative-fraction modeling, motivated by WAB-selective mismodeling in the selected background [6, Sec. 9.2]. Treating the two terms as independent gives

$$\delta_{f,2021}^{\mathrm{note}} = \sqrt{(0.035)^2 + (0.030)^2} = 0.046, \qquad f_{\mathrm{rad},2021}^{\mathrm{eff}} = 0.954\, f_{\mathrm{rad},2021}. \tag{12}$$

The 2015 and 2016 penalties remain at 7%.



(a) Full tridents, $\langle\sigma\rangle = 4.026 \times 10^9$ pb

(b) Radiative tridents, $\langle\sigma\rangle = 3.421 \times 10^7$ pb

(c) WAB, $\langle\sigma\rangle = 8.079 \times 10^{10}$ pb

Figure 16: Distribution of generated cross sections for background MC samples.

Figure 13: Generated cross-section distributions for the 2021 tritrig, radiative, and WAB background MC samples [6]. These distributions set the cross-section normalization inputs for the 2021 radiative-fraction calculation.

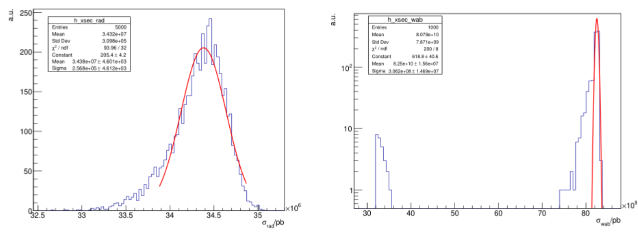


Figure 161: **Left:** Distribution of cross sections for radiative events, **Right:** Distribution of cross sections for WAB events.

Figure 14: RAD and WAB cross-section distributions used in the 2021 note's radiative-fraction cross-section uncertainty propagation [6]. Together with the relative normalization $R = 0.244$, these inputs lead to the 3.5% MC cross-section contribution.

The 4.6% term covers only the radiative-fraction normalization. The broader 2021 displaced-search uncertainty model in Fig. 15 also includes mass resolution, target position, and tight-selection effects that must be revisited before applying it to the full data set [6, Sec. 9, Table 58]. Those signal-yield and acceptance uncertainties are not included in Eq. (10).

Source	Percentage
MC Cross Section	3.5%
Pre-selection Uncertainty	3%
Invariant Mass Resolution	6.5%
Target Position Uncertainty	9%
MC-Data Tight Selection Uncertainty (Intuition)	2%
MC-Data Tight Selection Uncertainty (ANN)	4%

Table 58: Table of (expected) uncertainties for the displaced vertex search

Figure 15: Broader 2021 full-analysis-note systematic summary [6]. Only the MC cross-section and preselection entries are folded into the radiative-fraction-only penalty in Eq. (12); the other entries remain separate signal-yield or acceptance effects.

The selected spectra, resolution curves, and radiative fractions now specify the three data-dependent ingredients of the search: the observed mass distribution, the shape of a narrow signal, and the conversion from fitted signal yield to coupling. The next section describes how the GP predicts the smooth background beneath each tested mass.

4 Methodology

4.1 Background interpolation around a blinded mass window

At each mass hypothesis, the search must predict a smooth trident background without allowing a possible narrow resonance to influence that prediction. We therefore fit the GP on a fixed, dataset-specific mass range and exclude a small region around the test mass from training. The GP interpolates across the excluded region, and those same bins are then used in the signal-plus-background likelihood. Earlier HPS studies redefined the sidebands in separate piecewise windows. The fixed-support construction gives smoother behavior across the scan and allows the three data sets to enter a joint likelihood with a shared coupling.

4.2 Search interval and GP fit support

The search interval answers where a resonance hypothesis is tested. The GP fit support answers which surrounding bins constrain the smooth background at that hypothesis. Using the same endpoints for both definitions leaves the first and last test masses with a one-sided training sample, even though the input histogram contains usable data beyond the search. The three-campaign combined analysis uses the established 19–90, 39–180, and 50–250 MeV search intervals, with 14–135, 30–210, and 40–300 MeV fit supports for 2015, 2016, and 2021, respectively. The separate 2021-only analysis keeps the 50–250 MeV search interval and changes that dataset's fit support to 36–300 MeV.

The combined-analysis endpoints were fixed before its observed scan. The 2015 domain and the upper 2016 edge follow the earlier functional-form validation fits. The 2016 lower edge is placed at 30 MeV rather than 35 MeV because the exact rebin-five geometry matters: at a 39 MeV hypothesis the blind interval begins at 35.063 MeV, while the first bin center from a 35 MeV support is 35.125 MeV. A nominal 35–210 MeV support would therefore still contain no low-side training-bin center at the lower search endpoint. The adopted domains retain at least 12 and 20 low-side training bins at the 2015 and 2016 lower endpoints, and at least 138 and 51 high-side training bins at their upper endpoints. For the combined-analysis geometry, the corresponding 2021 counts are 9 and 55. We count the retained training bins explicitly at every active dataset and mass hypothesis rather than inferring them from continuous support boundaries.

For the 2021-only analysis, we compared lower support edges from 32 to 40 MeV using the same pseudoexperiments generated from a fitted 2021 source spectrum, with 30 MeV retained as a control. The comparison changes only the GP support, so differences in the extraction can be attributed to the location of the lower edge. Section 5.10 gives the selection and its limitations. It fixed the 2021-only support at 36–300 MeV, but does not change the 40–300 MeV support used in the three-campaign combination. The support study is conditional on its generating spectra and is not a direct coverage test. Table 3 keeps the two configurations distinct.

Analysis	2021 search interval	2021 GP support	Purpose and limitation
Three-campaign combination	50–250 MeV	40–300 MeV	Combines the 2015, 2016, and native-10% 2021 samples. This configuration is unchanged by the later 2021 support study.
2021-only analysis	50–250 MeV	36–300 MeV	Uses the native-10% 2021 sample after the conditional lower-edge study. The support choice is not a direct coverage result.

Table 3: The two analysis configurations that use the 2021 sample. Search interval and GP fit support are separate quantities; the combined and 2021-only analyses use different lower support edges.

4.3 Observable, scan grid, and blinding

The input observable is the reconstructed e^+e^- invariant-mass histogram for each dataset. Throughout this section, d labels the dataset, m denotes the reconstructed e^+e^- invariant mass, m_0 is the test-mass hypothesis, and $\sigma_d(m)$ is the dataset-specific mass resolution introduced in Section 3. The mass-hypothesis spacing is

$$\Delta m = 1\,\mathrm{MeV}, \tag{13}$$

and the blind window for a mass hypothesis m_0 is centered on the detector resolution,

$$\mathcal{B}_d(m_0) = \left[m_0 - n_{\mathrm{blind}}\sigma_d(m_0),\, m_0 + n_{\mathrm{blind}}\sigma_d(m_0)\right]. \tag{14}$$

The extraction half-width and GP training exclusion are independently configurable. Early injection studies used $n_{\mathrm{blind}} = 1.64$ with the same $1.64\,\sigma_d$ training exclusion. The nominal analysis uses a wider $2.25\,\sigma_d$ blind/extraction window with the GP training exclusion also set to $2.25\,\sigma_d$. For an ideal Gaussian line shape, the $1.64\,\sigma_d$ interval contains

$$f_{\mathrm{blind}}^{\mathrm{ideal}}(1.64) = \mathrm{erf}\left(\frac{1.64}{\sqrt{2}}\right) \simeq 0.899, \tag{15}$$

while the $2.25\,\sigma_d$ interval contains

$$f_{\mathrm{blind}}^{\mathrm{ideal}}(2.25) = \mathrm{erf}\left(\frac{2.25}{\sqrt{2}}\right) \simeq 0.976. \tag{16}$$

We normalize the signal over the full Gaussian line shape and use only its blinded-bin slice in the local likelihood; the slice is not renormalized inside the window.

The notation $\mathcal{B}_d(m_0)$ denotes the set of histogram bins that lie inside the blinded interval for dataset d at test mass m_0. In the nominal fit there is no additional gap between the extraction window and the GP training sample: when the extraction and training-exclusion half-widths are equal, the training exclusion starts immediately outside $\mathcal{B}_d(m_0)$. A wider exclusion, for example $1.96\,\sigma_d$ instead of $1.64\,\sigma_d$, reduces signal leakage into the training region but also increases the interpolation distance and therefore the GP predictive uncertainty. In the simple background-dominated proxy $S/\sqrt{B} \propto F(k)/\sqrt{k}$ with $F(k) = \mathrm{erf}(k/\sqrt{2})$, the optimum lies near $k \simeq 1.4$, so the earlier $1.64\,\sigma_d$ extraction was close to the pure-sensitivity optimum. Section 5 then motivates the current $2.25\,\sigma_d$

blind/extraction choice as a signal-recovery improvement after full-refit pseudoexperiments showed appreciable training-region absorption at narrower exclusions.

We assign bins by their fixed centers rather than splitting bins that overlap a boundary. Let $c_i = (e_i + e_{i+1})/2$ be the center of bin i. The full bin is included in the blinded likelihood when

$$m_0 - n_{\text{blind}}\sigma_d(m_0) \le c_i \le m_0 + n_{\text{blind}}\sigma_d(m_0), \tag{17}$$

and it enters the sideband sample only when its center lies strictly outside the training-exclusion interval. The same center mask is applied to the counts, GP prediction, covariance, and signal template, so no observed count is divided between the two regions.

The Gaussian signal template is integrated over the full bin edges in Eq. (33), but the selected set of bins changes discretely when a boundary passes a bin center. Consequently the quoted local significance and ε^2 limit are not meaningful to a precision finer than the mass binning and scan grid used to define these masks. The same discrete procedure is used for the data and pseudoexperiments.

4.4 Log-space preprocessing and the choice $\alpha = 1/y$

Following recent HEP GPR background-modeling studies [8, 9], we fit the GP in transformed coordinates:

$$x = \log m, \qquad z = \begin{cases} \log y, & y > 0, \\ 0, & y = 0, \end{cases} \tag{18}$$

where y is the observed bin count. The log transformation is used for two related reasons. First, the prompt trident spectrum spans a wide dynamic range, and modeling z rather than y turns multiplicative shape variations into approximately additive ones. Second, $x = \log m$ makes equal distances in the GP coordinate correspond to equal fractional changes in mass, which is the relevant scale once the detector resolution is written in the transformed coordinate as $\sigma_x(m) \simeq \sigma_d(m)/m$. Barr and Liu emphasize that applying logarithms to both axes improves convergence and performance for wide, steeply falling spectra [8]. The GP regression step uses scikit-learn's GaussianProcessRegressor estimator [10]; the surrounding HPS analysis code supplies the histogram handling, mass-dependent sideband masks, signal templates, profile likelihood, and $\mathrm{CL_s}$ limit construction.

The GP is therefore used to model an unobserved smooth mean function $z_{\text{true},d}(x)$ for the log-count spectrum. In that language, "latent" means only that the smooth log-rate is not observed directly; it is inferred from the histogram counts through the transformed coordinates and the heteroscedastic variance term. The GP mean vector is the vector of latent log-count means evaluated at the selected training-bin coordinate. For Poisson counts $Y_i \sim \text{Pois}(\lambda_i)$, the delta method gives $\text{Var}[\log Y_i] \approx \text{Var}[Y_i]/E[Y_i]^2 \approx 1/\lambda_i$, which motivates the plug-in choice $\alpha_i \approx 1/y_i$ for occupied bins. A heteroscedastic diagonal term α_i is then added to the training covariance,

$$\alpha_i = \begin{cases} 1/y_i, & y_i > 0, \\ 1, & y_i = 0, \end{cases} \tag{19}$$

$$K_{\text{obs}} = K(X, X) + \text{diag}(\boldsymbol{\alpha}), \tag{20}$$

so that each α_i plays the role of an observation-variance term for bin i. This is the quantity that GaussianProcessRegressor adds to the diagonal during fitting, so it is treated here as a variance model rather than an arbitrary tuning parameter.

Here α_i is the latent-space variance associated with the transformed counts; we do not add an extra $\log y$ weight. For empty bins the floor $\alpha_i = 1$ is a unit variance in log-count space, not a pseudo-count of one event. It simply prevents zero-count bins from receiving ill-defined or disproportionate weight after the log transformation. In this high-statistics search such bins are rare, so the floor acts mainly as a conservative numerical safeguard rather than as a dominant source of uncertainty.

The distinction from Ref. [8] is more specific than a difference in kernel choice. Barr and Liu also use a Constant×RBF kernel and optimize its amplitude and length scale. Their independent methodological study uses a spectrum derived from a published CMS search; it is not a CMS Collaboration GPR result. In addition to optimizing the kernel, they treat the diagonal array as tunable regularization. For occupied bins their nominal prescription is

$$\alpha_i^{\mathrm{BL}} = \frac{\sqrt{y_i}}{y_i \log y_i} = \frac{1}{\sqrt{y_i}\log y_i}, \tag{21}$$

and fixed regional multipliers f_i are then applied to α_i^{BL}. Reducing the first eleven multipliers to 0.1 raised the fraction of 100 background-only pseudoexperiments passing their BumpHunter criterion from 51% to 87%, but localized residual structures remained; their high-luminosity benchmark subsequently extended the reduced-multiplier region to the first twenty bins [8]. The study therefore demonstrates that diagonal tuning can improve a particular benchmark, while also showing that the tuned region is sample- and regime-dependent.

The active HPS configuration applies no regional multiplier to the early bins and fixes Eq. (19) as the delta-method variance model. This choice matters because scikit-learn's alpha parameter has both an observation-noise interpretation and a numerical Tikhonov-regularization interpretation [11]. HPS adopts the former and places its separately validated smoothness control in the allowed length-scale domain. These are complementary controls: changing α_i changes the leverage of bin i on the covariance diagonal, whereas changing ℓ changes the off-diagonal correlation range of the latent function. No matched study presented here establishes that one strategy is intrinsically superior to the other.

4.5 Kernel choice and distinct regularization roles

We model the background with a single ConstantKernel×RBF kernel,

$$k(x, x') = C \exp\left[-\frac{(x-x')^2}{2\ell^2}\right], \tag{22}$$

with broad amplitude bounds and a resolution-scaled RBF length-scale domain. This is the same simple kernel class used in the methodological studies of Refs. [9, 8], but the surrounding noise and validation choices need not be the same. The two quantities optimized in every HPS sideband fit are C and ℓ. The ConstantKernel value $C = k(x, x)$ is the pointwise prior variance of the latent log-rate, so $\sqrt{C}$ is its prior standard deviation; C is not a fitted event-count normalization. The length scale ℓ controls how quickly covariance falls with separation in $x = \log m$, and therefore how rapidly the background function may bend. The fixed α_i values are not a third kernel parameter and there is no optimized WhiteKernel in the baseline. They enter only through the observation-covariance diagonal.

Figure 16 separates these roles graphically. This separation is useful when discussing regularization: changing C rescales the covariance amplitude, changing ℓ changes the correlation geometry and smoothness of the latent function, and changing α_i changes how strongly one observation constrains that function.

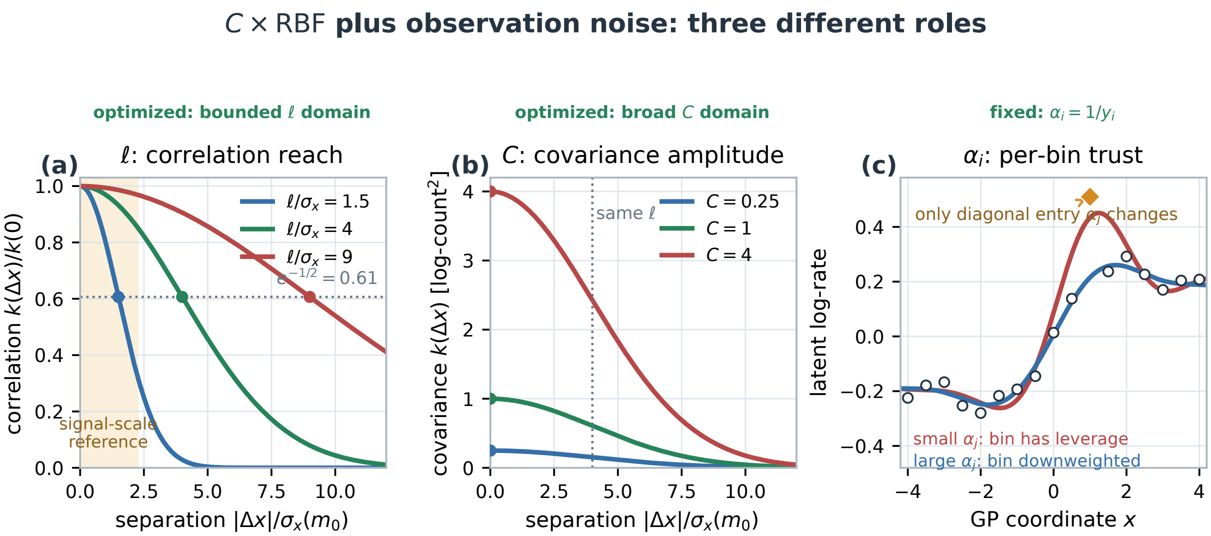


Figure 16: Roles of the three quantities in the GPR covariance. Panel (a) shows the normalized RBF correlation for three illustrative values of $\ell/\sigma_x(m_0)$; at a separation equal to ℓ, every curve has correlation $e^{-1/2} \simeq 0.61$. The length scale sets the correlation range. Panel (b) varies C at fixed ℓ. It changes the covariance amplitude, with pointwise prior standard deviation $\sqrt{C}$, without changing the normalized correlation range. Panel (c) holds the kernel fixed and changes one diagonal variance α_j, thereby changing that bin's statistical weight. We optimize C and ℓ on the sidebands and fix α_i from the count-based variance rule. The curves are schematic.

We use the Constant×RBF model. A Constant×Matérn kernel with $\nu = 3/2$ and piecewise kernels remain exploratory robustness checks; they have not been tested for coverage in the three-campaign configuration.

4.6 Resolution-scaled length-scale parameterization and bounds

Because the GP is trained in $x = \log m$, detector resolution must be translated into log-mass units. We define

$$\sigma_x(m) = \log(m + \sigma_d(m)) - \log m \simeq \frac{\sigma_d(m)}{m} \qquad (\sigma_d \ll m), \tag{23}$$

and use this local physical scale to set the RBF bounds. Under the local, resolution-scaled policy,

$$\ell_{\min,d}(m) = k_{\min,d}\sigma_x(m), \qquad \ell_{\max,d}(m) = k_{\max,d}\sigma_x(m). \tag{24}$$

For the RBF covariance used here,

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \exp\left[-\frac{(x - x')^2}{2\ell^2}\right], \qquad \frac{k_{\text{RBF}}(x, x + \ell)}{k_{\text{RBF}}(x, x)} = e^{-1/2} \simeq 0.61. \tag{25}$$

Thus ℓ is a covariance-decay scale in the GP input coordinate: larger values favor smoother background functions, while smaller values admit more rapid variation [12, 13]. The ratio $\ell/\sigma_x(m)$ expresses that nuisance-model smoothness relative to the width of the narrow signal hypothesis. It is not, without an independently validated mapping, a detector resolution measurement or a physical background correlation length. The fitted ℓ_{opt} is the log-marginal-likelihood optimum conditional on the kernel, data support, sideband mask, noise model, and configured optimizer domain. Consequently an optimum at $\ell_{\max}$ establishes only that the numerical constraint is active; it neither measures a physical scale at the boundary nor locates the unconstrained optimum. The implementation optimizes log-transformed kernel hyperparameters within these configured bounds [11].

For a fixed dataset d and mass hypothesis m_0, let $\boldsymbol{z}_{\mathrm{sb}}$ contain only the transformed sideband observations and define

$$K_y(C, \ell) = C\, R_\ell(X_{\mathrm{sb}}, X_{\mathrm{sb}}) + \mathrm{diag}(\boldsymbol{\alpha}_{\mathrm{sb}}). \tag{26}$$

The optimized pair maximizes

$$\log p(\boldsymbol{z}_{\mathrm{sb}} \mid X_{\mathrm{sb}}, C, \ell) = -\frac{1}{2}\boldsymbol{z}_{\mathrm{sb}}^T K_y^{-1} \boldsymbol{z}_{\mathrm{sb}} - \frac{1}{2}\log|K_y| - \frac{n_{\mathrm{sb}}}{2}\log(2\pi). \tag{27}$$

The quadratic term rewards agreement with the sideband observations, while the determinant term accounts for the covariance volume used to obtain that agreement. The implementation searches in the logarithms of C and ℓ with multiple optimizer restarts. This is an empirical-Bayes, or type-II maximum-likelihood, selection of the covariance parameters; it is not a profile of ℓ as a frequentist nuisance parameter in the subsequent local $\mathrm{CL_s}$ likelihood.

Figure 17 shows why the optimum is specific to a dataset and mass hypothesis. Moving m_0 changes which bins are excluded from training and changes the resolution-scaled domain through $\sigma_x(m_0)$. The same dataset can therefore return a different $(C_{\mathrm{opt}}, \ell_{\mathrm{opt}})$ at neighboring hypotheses even though the kernel family is unchanged. An interior maximum means only that the configured range is nonbinding for that fit. A maximum on a boundary means that the range is active, while a long shallow likelihood ridge means that the parameters are weakly identified. None of these cases turns ℓ_{opt} into a direct measurement of detector resolution or a physical background correlation length.

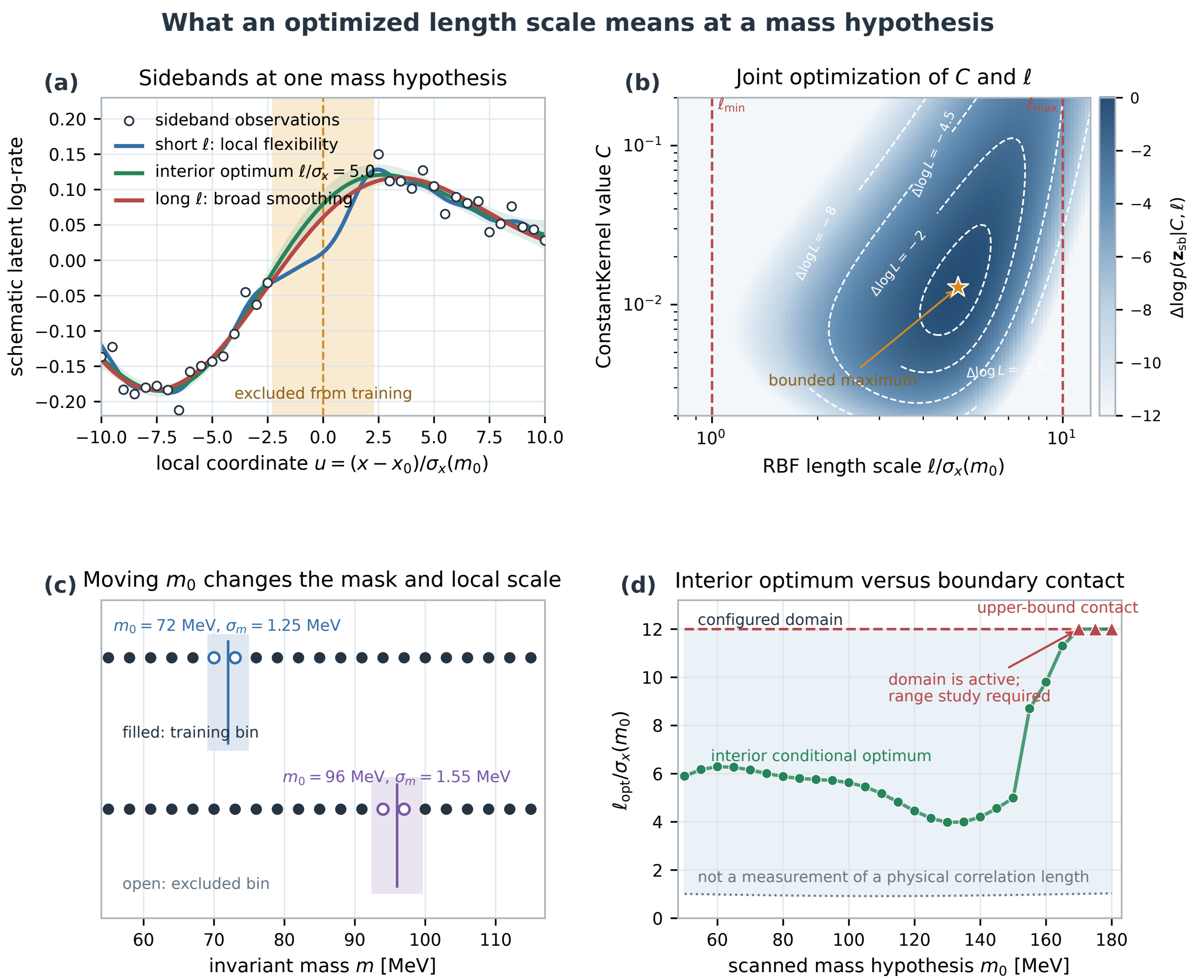


Figure 17: Meaning of per-mass-hypothesis hyperparameter optimization. Panel (a) shows one schematic sideband problem in $u = (x - x_0)/\sigma_x(m_0)$. The shaded $2.25\,\sigma_x$ interval is excluded from training; the three curves illustrate the different interpolation behavior allowed by short, interior, and long candidate length scales. Panel (b) shows the joint log-marginal-likelihood surface for the same synthetic sidebands. The star is the bounded maximum in (C, ℓ), and the elongated contours expose their possible tradeoff. Panel (c) shows that moving m_0 changes both the sideband mask and the local resolution scale that defines the allowed ℓ interval. Panel (d) illustrates how a scan records an interior optimum differently from repeated upper-bound contact: the former is a conditional best fit within a nonbinding range, whereas the latter diagnoses an active numerical domain and motivates a predeclared range and closure study. The figure is explanatory only; it contains no HPS observations or fit results.

This distinction determines how the hyperparameter study enters a physics result. If the allowed range forces ℓ too low, the GP retains unnecessary signal-scale flexibility and can absorb part of a localized excess; if the fitted background is forced too smooth, broad mismodeling can instead leave a biased local residual. Raising only the upper bound permits, but does not require, a smoother solution. A larger data set also need not prefer a larger ℓ: additional statistics can resolve background structure and move the optimum downward. The upper range is therefore selected from boundary occupancy, likelihood ordering, optimizer stability, and fixed-amplitude signal-recovery diagnostics, never from a favorable observed limit or local p_0. These checks can justify a bounded candidate range, but they do not by themselves establish frequentist coverage.

The three-campaign analysis uses the dataset-specific factors in Table 4. The lower factors come from full-refit closure studies; changing the 2016 upper factor does not requalify those studies at full statistics.

Dataset	$k_{\min,d}$	$k_{\max,d}$	Basis
2015 full	1.0	8	full-refit closure study
2016 full	0.9	12	closure plus post-selection range study
2021 10%	1.1	15	matched-refit closure study

Table 4: Resolution-scaled RBF length-scale factors used in the three-campaign analysis. The factors multiply $\sigma_x(m)$ in Eq. (24). The 2016 upper factor was chosen after a range limitation appeared in the observed scan; it is therefore a post-selection choice, not a pre-unblinding or coverage qualification.

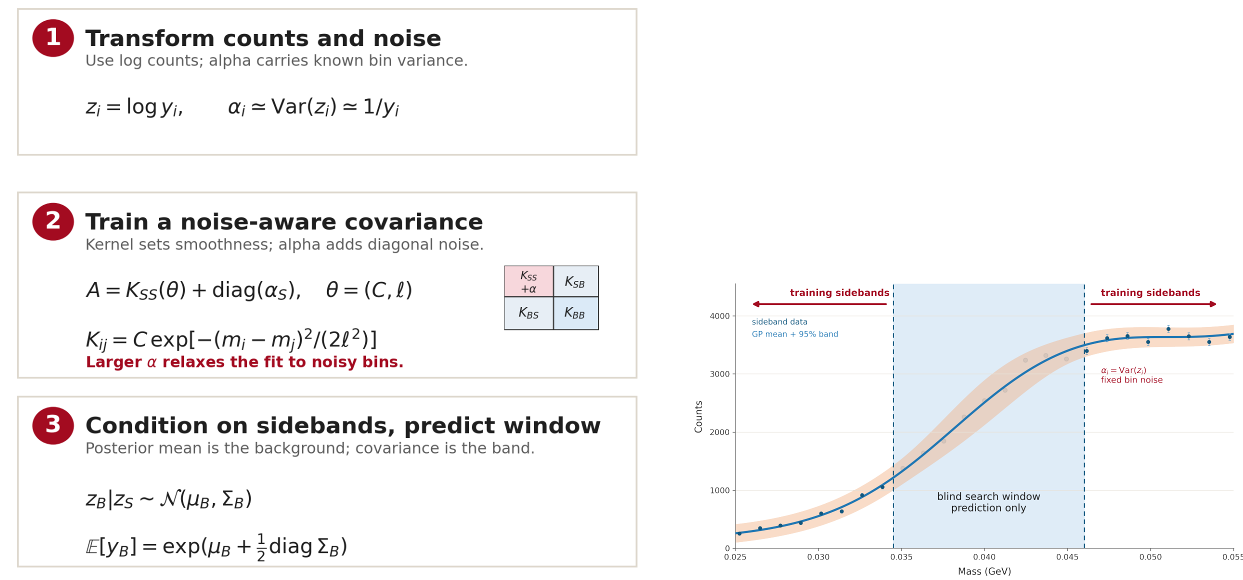
The lower factors are based on full-refit functional-form closure scans, with the observed-spectrum result kept out of the selection criterion. For 2015, the 100-toy scan slightly prefers $k_{\min} = 1.1$ in its aggregate score, but $k_{\min} = 1.0$ has similar pull-width calibration and a weaker expected-reach proxy. We retain 1.0 as the more conservative choice. For 2016, the 25-toy continuation supports the change to $k_{\min} = 0.9$: its nonzero-injection RMS pull mean is 0.321 and its median pull width is 0.948, compared with 0.403 and 0.889 for $k_{\min} = 1.1$. It gives the smallest aggregate calibration score among the scanned 0.9, 1.0, and 1.1 rows and reduces the coherent apparent signal-recovery bias that motivated the follow-up. For 2021, the corrected matched-reference scan finds small rank separations among the same three settings. The $k_{\min} = 1.1$ row has the median pull width closest to unity and the best extraction-level ε^2 proxy, while the 0.9 and 1.0 rows have slightly smaller aggregate scores. We choose 1.1 as a calibration–reach tradeoff, not because it wins every diagnostic. These studies used the data fractions stated in Section 5 and do not establish full-statistics closure.

The initial upper factors were 8, 8, and 15 for 2015, 2016, and 2021. In the reviewed observed scan, the 2016 optimum reached its upper bound at all 142 hypotheses. We therefore varied only $k_{\max,2016}$, evaluating factors 8, 10, 12, 15, and 20 on the same 39–180 MeV mass grid. A fit is counted as upper-bound occupied when $\ell_{\rm opt}/\ell_{\max} \geq 0.999$. The respective occupancies are 142, 56, 0, 0, and 0 of 142. Factor 12 is the first scanned setting with no occupied hypotheses. It is also followed by a numerical plateau: from factor 12 to 15, the largest pointwise change in the 2016 observed yield upper limit is 0.083%, the largest $|\Delta Z|$ is 0.00255, and the log-marginal-likelihood differences lie between -4.5×10^{-6} and 8.0×10^{-6}. The factor-15 to factor-20 comparison is similarly stable. The selection is based on removal of the artificial boundary and the next-setting plateau, not on which observed limit is tighter or which local p_0 is smaller.

Because the observed spectrum revealed the range limitation, factor 12 is a post-selection choice for the three-campaign analysis. It is not a pre-unblinding choice or a coverage-calibrated setting; full-statistics functional-form closure and direct coverage remain separate questions. The configurations continue to retain a dataset-wide median-resolution floor on $\ell_{\max,d}$ so that the ceiling does not become artificially small where the local resolution is exceptionally tight. This is a guard based on a representative σ_x value, not a floor derived from luminosity or event statistics. Alternative upper bounds, fixed-background fits, and the common-lower-bound overlays remain robustness studies collected in the appendices.

An initial 2021 source/exposure study used a nested-Poisson construction. It motivated the later matched-refit studies but is not part of the procedure used here.

Figure 18 summarizes the same construction in the language used by the implemented likelihood. The sideband bins set the log-count training sample and its heteroscedastic bin-noise assignment, the kernel hyperparameters are optimized only on those sidebands, and the trained GP is then conditioned to predict the mean and covariance in the blinded mass window. That predicted count-space mean and covariance are the background inputs passed to the local profiled signal extraction.



(a) Log-count transform, noise-aware covariance training, and window prediction. The filled covariance boxes indicate the sideband-sideband, sideband-blind, and blind-blind kernel blocks used when conditioning the GP.

(b) Sideband-only training and blinded-window prediction. Filled markers show binned count data, vertical error bars show the fixed per-bin statistical noise, and the horizontal reference lines are count-axis guides used to compare sideband and blinded-window scales.

Figure 18: Kernel and blinded-window construction. The GP is trained on sideband bins with fixed bin-noise variances, not on the blinded bins. The posterior prediction in the blind window supplies the background mean and covariance that enter the local likelihood; the shaded band in the schematic denotes the GP posterior interval and the dashed vertical lines mark the blinded-window boundaries.

4.7 Blind-window prediction in count space

For each mass point, we train the GP only on the bins outside the training exclusion window and predicts the posterior mean and covariance of the latent smooth log-rate in the blinded bins. Since the fit is performed in log-count space, that posterior is mapped back to count space through the standard lognormal moment relations:

$$\mu_i = \exp\left(\mu_i^{(\log)} + \frac{1}{2}\Sigma_{ii}^{(\log)}\right), \tag{28}$$

$$\mathrm{Cov}(\lambda_i, \lambda_j) = \mu_i \mu_j \left[\exp\left(\Sigma_{ij}^{(\log)}\right) - 1\right]. \tag{29}$$

The resulting blind-window mean vector $\boldsymbol{b}_d(m)$ and covariance matrix $\boldsymbol{C}_d(m)$ enter the local extraction fit and the $\mathrm{CL_s}$ limit. The Poisson fluctuation of the observed $n_{d,i}$ is handled separately by the likelihood, so $\boldsymbol{C}_d$ should be read as the covariance of the GP-predicted background mean rather than the covariance of the observed histogram counts themselves.

To propagate that correlated GP uncertainty into the profile likelihood, the analysis uses the Cholesky factorization

$$\boldsymbol{C}_d(m) = L_d(m)L_d^\top(m), \tag{30}$$

with $L_d(m)$ the lower-triangular Cholesky factor of the positive-semidefinite blind-window covariance. Introducing an auxiliary standard-normal nuisance vector $\boldsymbol{\theta}_d \sim \mathcal{N}(0, I)$ and the background-deformation field $\Delta\boldsymbol{b}_d(m) = L_d(m)\boldsymbol{\theta}_d$ then gives $\Delta\boldsymbol{b}_d \sim \mathcal{N}(0, \boldsymbol{C}_d)$. This is the standard HEP construction for encoding correlated Gaussian-constrained nuisance variations inside a profile likelihood model [14, 15]. In the present analysis it is especially important because the GP prediction induces nontrivial bin-to-bin correlations across the blinded window; carrying those correlations through $L_d\boldsymbol{\theta}_d$ lets the local significance and $\mathrm{CL_s}$ limit respond to coherent background deformations rather than misinterpreting the GP uncertainty as independent per-bin noise.

4.8 Signal model and conversion to ε^2

For each dataset d and mass hypothesis m_0, we first construct a full-range Gaussian line-shape template over the histogram binning,

$$w_{d,i}^{\mathrm{full}}(m_0) = \int_{e_i}^{e_{i+1}} \frac{1}{\sqrt{2\pi}\sigma_d(m_0)} \exp\left[-\frac{(m-m_0)^2}{2\sigma_d(m_0)^2}\right] dm, \tag{31}$$

where e_i and e_{i+1} are the lower and upper edges of histogram bin i. The template is normalized so that

$$\sum_{i\in\mathrm{full\ range}} w_{d,i}^{\mathrm{full}}(m_0) = 1. \tag{32}$$

The template entering the local likelihood is then the blinded-bin slice of this full-range object,

$$w_{d,i}(m_0) = \begin{cases} w_{d,i}^{\mathrm{full}}(m_0), & i \in \mathcal{B}_d(m_0), \\ \text{not used in the local fit}, & i \notin \mathcal{B}_d(m_0). \end{cases} \tag{33}$$

By the blinded-bin slice we mean exactly this restriction of the full normalized Gaussian template to the subset of bins in $\mathcal{B}_d(m_0)$; the template is not renormalized after the restriction. Its blinded-bin sum is therefore

$$f_{\mathrm{blind},d}(m_0) = \sum_{i\in\mathcal{B}_d(m_0)} w_{d,i}(m_0) < 1, \tag{34}$$

with a value near 0.90 for a $1.64\,\sigma_d$ half-width and near 0.98 for the current $2.25\,\sigma_d$ observed-limit half-width. The signal amplitude A_d is defined as the *total* local Gaussian signal yield associated with the mass hypothesis, not merely the yield inside the blind window. The expected signal contribution entering bin i of the local likelihood is therefore

$$s_{d,i}(A_d; m_0) = A_d\, w_{d,i}(m_0), \tag{35}$$

so the total expected signal actually visible inside the blind window is $f_{\mathrm{blind},d}(m_0)\, A_d$.

This convention preserves the fact that a finite blind region never contains the entire Gaussian line shape. It also keeps the signal parameter used in the fit aligned with the full-histogram injection model used in the procedural toy studies.

The conversion between signal yield and dark-photon coupling follows the standard fixed-target relation between A' production and radiative trident production [16, 5, 2]. In the present notation,

$$\left.\frac{dN_{A'}}{dm}\right|_{m=m_{A'}} = \frac{3\pi}{2N_{\text{eff}}\alpha_{\text{EM}}}\, m_{A'}\,\varepsilon^2 \left.\frac{dN_{\gamma^*}}{dm}\right|_{m=m_{A'}}, \tag{36}$$

where α_{EM} is the fine-structure constant, the factor $3\pi/2$ is the standard narrow-width coefficient relating on-shell A' production to radiative trident production, and N_{eff} counts the number of kinematically open visible decay channels contributing to the total A' width. Below the dimuon threshold, only the e^+e^- channel is open and the minimal-model conversion has $N_{\text{eff}} = 1$. The present GPR implementation keeps this e^+e^- normalization as the analysis convention for the prompt electron-channel yield; minimal dark-photon interpretations above $2m_\mu$ require updating the visible branching factor to include the newly open $\mu^+\mu^-$ channel. Writing the radiative differential rate as

$$\frac{dN_{\gamma^*}}{dm} = f_{\text{rad},d}(m)\,\rho_d(m), \tag{37}$$

where $\rho_d(m)$ is the differential prompt background rate in the selected invariant-mass spectrum, we convert between A_d and ε^2 through

$$\varepsilon^2 = \frac{2\alpha_{\text{EM}}\,A_d}{3\pi\, m\, f_{\text{rad},d}(m)\,\rho_d(m)}, \tag{38}$$

where $\rho_d(m)$ is estimated from the observed histogram as a local counts-per-GeV density in the same mass interval used for the nominal blind window,

$$\rho_d(m) \equiv \frac{N_d[m - n_{\text{blind}}\sigma_d(m),\, m + n_{\text{blind}}\sigma_d(m)]}{2\, n_{\text{blind}}\sigma_d(m)}, \tag{39}$$

with $n_{\text{blind}} = 1.64$ retained as the current prompt-density normalization window even when the extraction blind window is widened to $2.25\,\sigma_d$. This keeps the yield-to-ε^2 conversion tied to the established local density scale while the likelihood itself can recover the wider signal tails. The density-window width remains configurable so that alternative conventions can be reproduced when needed. It is convenient to define the dataset-specific conversion factor

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi\, m\, f_{\text{rad},d}(m)\,\rho_d(m)}{2\alpha_{\text{EM}}}, \tag{40}$$

so that a shared coupling ε^2 predicts a dataset-specific total signal yield $A_d = K_d(m)\varepsilon^2$.

Unless stated otherwise, the methodology section uses the nominal radiative fraction $f_{\text{rad},d}(m)$. The optional penalty that propagates normalization uncertainty into ε^2 is treated as a systematic variation rather than as part of the core GP fit and is discussed separately in Section 3.6. The numerical penalty values are therefore result-configuration choices, not ingredients of the background interpolation or local extraction fit. When that systematic option is enabled for dataset d, the effective radiative fraction is

$$f^{\text{eff}}_{\text{rad},d}(m) = (1 - \delta_{f,d})\, f_{\text{rad},d}(m), \tag{41}$$

so the same signal-yield limit maps to a weaker ε^2 limit.

4.9 Statistical inference in a single dataset

For a fixed dataset d and mass hypothesis m, let $\boldsymbol{n}_d$ denote the vector of observed blinded-bin counts, $\boldsymbol{b}_d$ the GP-predicted blinded-bin mean, $\boldsymbol{C}_d = L_d L_d^\top$ the corresponding covariance of Section 4.7 and Eq. (30), and $\boldsymbol{w}_d(m)$ the non-renormalized blinded-bin slice of the full Gaussian template defined in Eq. (33). The profiled local likelihood is

$$\mathcal{L}_d(A_d, \boldsymbol{\theta}_d) = \left[\prod_{i=1}^{N_d(m)} \mathrm{Pois}(n_{d,i} \,|\, b_{d,i} + (L_d\boldsymbol{\theta}_d)_i + A_d w_{d,i}(m))\right] \exp\!\left(-\frac{1}{2}\boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\right), \tag{42}$$

where the Gaussian nuisance vector $\boldsymbol{\theta}_d \sim \mathcal{N}(0, I)$ induces the correlated background-deformation field $L_d\boldsymbol{\theta}_d$ defined below Eq. (30), and $N_d(m)$ is the number of bins in the local blinded set $\mathcal{B}_d(m)$. The signal contribution entering the Poisson mean is exactly the blinded-bin restriction of Eq. (35).

At fixed m, profiling Eq. (42) with no physical sign constraint on A_d yields the unconstrained extraction estimator $\hat{A}_d^{\mathrm{unc}}$ and its local uncertainty $\sigma_{A,d}$. This unconstrained fit is used for closure, linearity, and pull studies because it allows downward background fluctuations to be captured by the estimator itself rather than being clipped at zero. For discovery and For upper limits, however, the parameter of interest is restricted to the physical region $A_d \geq 0$, and the corresponding physical best fit is

$$\hat{A}_d = \max\!\left(0, \hat{A}_d^{\mathrm{unc}}\right). \tag{43}$$

We denote by $\hat{\boldsymbol{\theta}}_d$ the nuisance-parameter maximizer at the unconditional best fit and by $\hat{\hat{\boldsymbol{\theta}}}_d(A_d)$ the conditional maximum likelihood estimator of $\boldsymbol{\theta}_d$ for fixed A_d.

4.9.1 Local significance

The single-dataset discovery statistic is the bounded profile-likelihood ratio

$$q_{0,d}(m) = -2\ln \frac{\mathcal{L}_d\!\left(A_d = 0, \hat{\hat{\boldsymbol{\theta}}}_d(0)\right)}{\mathcal{L}_d\!\left(\hat{A}_d, \hat{\boldsymbol{\theta}}_d\right)}, \tag{44}$$

with $\hat{A}_d \geq 0$. The local one-sided discovery p-value and the corresponding Gaussian significance are then obtained with the asymptotic Cowan-et-al. mapping [15]. The resulting local-significance definition is explicitly one-sided and discovery facing: downward fluctuations are allowed in the unconstrained fit used for closure studies, but they do not generate a positive local discovery significance. We use this profiled multi-bin test throughout.

4.9.2 CL$_\mathrm{s}$ upper limits

To test a fixed non-negative signal amplitude A_d, the analysis uses the bounded one-sided profile-likelihood statistic

$$\tilde{q}_{A_d}(m) = \begin{cases} 0, & \hat{A}_d^{\mathrm{unc}} > A_d, \\ -2\ln \dfrac{\mathcal{L}_d\!\left(A_d, \hat{\hat{\boldsymbol{\theta}}}_d(A_d)\right)}{\mathcal{L}_d\!\left(\hat{A}_d^{\mathrm{unc}}, \hat{\boldsymbol{\theta}}_d\right)}, & 0 \leq \hat{A}_d^{\mathrm{unc}} \leq A_d, \\ -2\ln \dfrac{\mathcal{L}_d\!\left(A_d, \hat{\hat{\boldsymbol{\theta}}}_d(A_d)\right)}{\mathcal{L}_d\!\left(0, \hat{\hat{\boldsymbol{\theta}}}_d(0)\right)}, & \hat{A}_d^{\mathrm{unc}} < 0, \end{cases} \tag{45}$$

which is the standard bounded Cowan construction for upper limits [15]. The corresponding modified-frequentist tail areas are

$$\mathrm{CL}_{s+b}(A_d;m) = P\Big(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\mathrm{obs}} \Big| A_d + \mathrm{bkg}\Big), \qquad \mathrm{CL}_b(A_d;m) = P\Big(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\mathrm{obs}} \Big| \mathrm{bkg}\Big), \tag{46}$$

and

$$\mathrm{CL}_s(A_d;m) = \frac{\mathrm{CL}_{s+b}(A_d;m)}{\mathrm{CL}_b(A_d;m)}. \tag{47}$$

Dividing by CL_b prevents a downward background fluctuation from producing an artificially aggressive exclusion in a region with little intrinsic sensitivity [17, 18]. At confidence level $1-\alpha$, the single-dataset upper limit is

$$A_{\mathrm{UL},d}(m;\alpha) = \min\{A_d : \mathrm{CL}_s(A_d;m) \leq \alpha\}. \tag{48}$$

The analysis uses $\alpha = 0.10$, corresponding to a 90% CL upper limit; comparison studies at 95% use $\alpha = 0.05$. We report the limit both in signal-yield space and, after applying the dataset-specific conversion factor of Eq. (40), in ε^2 space.

4.10 Asymptotic and toy calibration

The local-significance mapping introduced above is treated asymptotically throughout the scan. For upper limits, the same bounded statistic $\tilde{q}_{A_d}$ is calibrated in one of two ways. In asymptotic mode, the background-only Asimov data set is used to evaluate the reference quantity $\tilde{q}_{A_d,A}$, so that

$$\mathrm{CL}_{s+b}(A_d;m) = 1 - \Phi\Big(\sqrt{\tilde{q}_{A_d}^{\mathrm{obs}}}\Big), \qquad \mathrm{CL}_b(A_d;m) = \Phi\Big(\sqrt{\tilde{q}_{A_d,A}} - \sqrt{\tilde{q}_{A_d}^{\mathrm{obs}}}\Big). \tag{49}$$

In toy mode, the same statistic $\tilde{q}_{A_d}$ is calibrated directly with background-only and signal-plus-background pseudoexperiments rather than by switching to a different ordering variable. The shared-coupling combination described below uses the identical logic after replacing the single-dataset amplitude A_d with the common parameter of interest ε^2.

The pseudoexperiment program remains important even with the asymptotic formulae in place because the HPS likelihood combines a multi-bin Poisson term, a GP-conditioned blind-window covariance, a non-renormalized Gaussian signal template, and a mass-dependent yield-to-ε^2 conversion. The pull, spurious-signal, and injection-recovery diagnostics of Section 5 test this machinery under the declared conditional backgrounds. They are not direct $\mathrm{CL_s}$ coverage studies or a scan-wide calibration of Z.

4.11 Matched-reference injection–extraction diagnostics

We define the injection scale with a background-only reference fit that uses the same support, binning, exclusion mask, kernel bounds, preprocessing, and GP refit as the corresponding pseudo-experiment. If $\mathcal{F}[\boldsymbol{n};\boldsymbol{w}]$ denotes that full procedure for counts $\boldsymbol{n}$ and the blind-window slice of the unit-normalized full-range signal template $\boldsymbol{w}$, then

$$(\widehat{A}_{\mathrm{ref}}, \sigma_{A,\mathrm{ref}}) = \mathcal{F}[\boldsymbol{n}_{\mathrm{ref}};\boldsymbol{w}], \qquad A_{\mathrm{inj}}(z) = z\,\sigma_{A,\mathrm{ref}}, \qquad z \in \{0,1,3,5\}. \tag{50}$$

The reference uncertainty is a fixed calibration of the injection coordinate at a given source, toy index, and mass. It is not the post-injection uncertainty returned for an individual extraction.

For pseudoexperiment t, the signed pull and paired response are

$$p_t(z) = \frac{\widehat{A}_t(z) - A_{\mathrm{inj},t}(z)}{\sigma_{A,t}(z)}, \tag{51}$$

$$R_t(z) = \frac{\widehat{A}_t(z) - \widehat{A}_t(0)}{A_{\mathrm{inj},t}(z)}, \qquad z > 0. \tag{52}$$

The zero-injection mean pull diagnoses a conditional spurious-signal offset under the declared generating truth. Its width probes the returned uncertainty scale, while R_t separates injection response from a background-specific zero-signal offset. These quantities use the toy-level $\sigma_{A,t}$ in the denominator. A finite 25-background sample supports two-sided 90% Student-t intervals on the mean and descriptive widths, not a direct coverage statement.

Full refitting and fixed-background displays answer different questions. Holding the GP fixed isolates the local profile-likelihood extraction. Refitting the GP on every pseudo-spectrum additionally tests background adaptation and possible signal absorption. We use the latter matched-refit procedure. Numerical acceptance is decided from likelihood, reproducibility, covariance, and kernel-coordinate diagnostics before amplitudes or pulls are summarized.

4.12 Observed-limit toy diagnostics

The analysis uses two distinct families of p-values for different purposes. The first is the discovery statistic of Eq. (44), summarized through $q_0(m)$, $p_0(m)$, and $Z_{\mathrm{local}}(m)$. The second consists of diagnostics derived from the background-only $\mathrm{CL_s}$ limit ensemble at each mass. For N_{toys} background-only pseudoexperiments and an observed limit $\varepsilon^2_{\mathrm{UL,obs}}(m)$ at the confidence level under study, those quantities are

$$p_{\mathrm{strong}}(m) = \frac{1}{N_{\mathrm{toys}}} \sum_{t=1}^{N_{\mathrm{toys}}} \mathbf{1}\left[\varepsilon^2_{\mathrm{UL},t}(m) \le \varepsilon^2_{\mathrm{UL,obs}}(m)\right], \tag{53}$$

$$p_{\mathrm{weak}}(m) = \frac{1}{N_{\mathrm{toys}}} \sum_{t=1}^{N_{\mathrm{toys}}} \mathbf{1}\left[\varepsilon^2_{\mathrm{UL},t}(m) \ge \varepsilon^2_{\mathrm{UL,obs}}(m)\right], \tag{54}$$

$$p_{\mathrm{two}}(m) = \min[1,\, 2\min(p_{\mathrm{strong}}(m),\, p_{\mathrm{weak}}(m))]\,. \tag{55}$$

Here p_{strong} quantifies how unusually *strong* the observed limit is relative to the background-only toy ensemble, while p_{weak} quantifies how unusually *weak* it is. The two-sided quantity p_{two} is retained only as a symmetric diagnostic; because it responds to both unusually strong and unusually weak limits, it is less targeted than the one-sided discovery or exclusion tests and is not used as a search statistic. Figure 19 gives the compact fixed-mass definition of these quantities, while Figure 20 shows representative strong-limit, weak-limit, and bounded median-case examples from the same toy-limit logic.

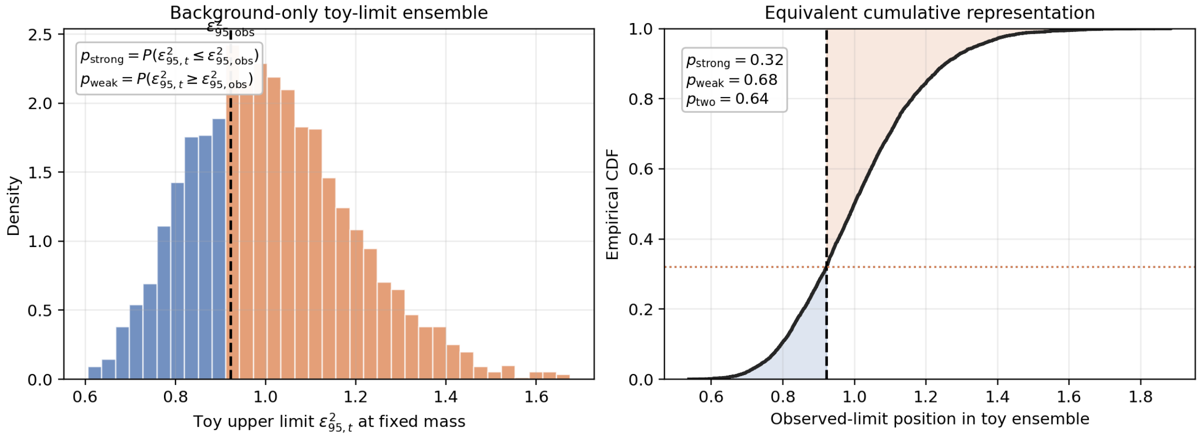


Figure 19: Illustrative toy-limit diagnostic at fixed mass. The left panel shows a background-only ensemble of observed $\mathrm{CL_s}$ upper limits, and the right panel shows the corresponding empirical cumulative distribution. The vertical line marks the observed limit. The lower-tail area defines p_{strong}, the upper-tail area defines p_{weak}, and $p_{\mathrm{two}} = \min[1, 2\min(p_{\mathrm{strong}}, p_{\mathrm{weak}})]$ is used only as a symmetric diagnostic summary.

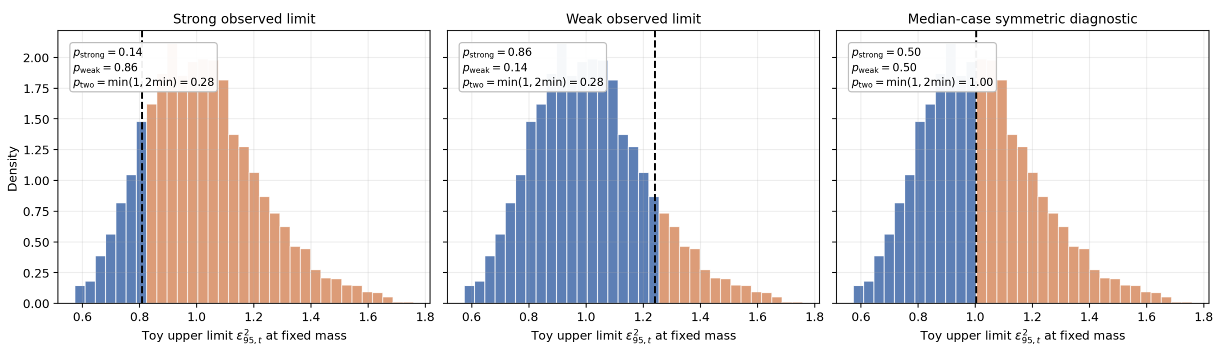


Figure 20: Representative fixed-mass observed-limit toy diagnostics. The left panel shows an unusually strong observed limit relative to the background-only toy ensemble, the middle panel shows an unusually weak observed limit, and the right panel shows a near-median case where the bounded symmetric summary saturates at $p_{\mathrm{two}} = 1$. These are limit diagnostics, not discovery statistics.

These quantities are therefore limit diagnostics rather than discovery-significance measures in the sense of p_0. If the observed limit lies near the median of the background-only ensemble, then roughly half of the toys lie above it and half lie below it, making both p_{strong} and p_{weak} naturally $\mathcal{O}(0.5)$. Their finite resolution is $1/N_{\mathrm{toys}}$. In the three-campaign combined ensemble, $N_{\mathrm{toys}} = 300$, so a one-count change is $1/300 \simeq 0.0033$ and a zero-count tail is reported as 0 of 300 rather than as an exact zero probability. These toy-limit diagnostics cannot by themselves establish or refute a high-significance local excess in the discovery statistic defined by the one-sided local p_0 mapping above.

4.13 Shared-ε^2 combination across datasets

At a common mass hypothesis m, each active dataset contributes a distinct blinded-bin count vector, GP background prediction, covariance matrix, and Gaussian signal template. Let the set of datasets that cover the hypothesis be $\mathcal{D}(m)$. For each $d \in \mathcal{D}(m)$, define

$$\boldsymbol{n}_d(m), \qquad \boldsymbol{b}_d(m), \qquad \boldsymbol{C}_d(m) = L_d L_d^\top, \qquad \boldsymbol{w}_d(m), \qquad K_d(m), \tag{56}$$

where $\boldsymbol{w}_d(m)$ is the blinded-bin template slice of Eq. (33), $K_d(m)$ is the yield-per-unit-ε^2 conversion of Eq. (40), and L_d is the Cholesky factor of the GP covariance in that dataset. When the radiative-fraction penalty of Eq. (10) is enabled, the conversion becomes

$$K_d^{\text{eff}}(m) = (1 - \delta_{f,d})\, K_d(m), \tag{57}$$

so the expected signal vector per unit ε^2 is reduced coherently in every active dataset. The per-dataset likelihood is then

$$\mathcal{L}_d(\varepsilon^2, \boldsymbol{\theta}_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}\Big(n_{d,i} \,\Big|\, b_{d,i} + (L_d\boldsymbol{\theta}_d)_i + \varepsilon^2 K_d^{\text{eff}}(m) w_{d,i}(m)\Big)\right] \exp\Big(-\frac{1}{2}\boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\Big). \tag{58}$$

The baseline combined likelihood is the product over datasets,

$$\mathcal{L}_{\text{comb}}(\varepsilon^2, \{\boldsymbol{\theta}_d\}) = \prod_{d \in \mathcal{D}(m)} \mathcal{L}_d(\varepsilon^2, \boldsymbol{\theta}_d). \tag{59}$$

Equivalently, one may concatenate all blinded-bin vectors into a single object. In the following equations, $\|$ denotes vector concatenation, e.g. $\boldsymbol{n}_1 \,\|\, \boldsymbol{n}_2$:

$$\boldsymbol{n} = \boldsymbol{n}_1 \,\|\, \boldsymbol{n}_2 \,\|\, \cdots, \tag{60}$$

$$\boldsymbol{b} = \boldsymbol{b}_1 \,\|\, \boldsymbol{b}_2 \,\|\, \cdots, \tag{61}$$

$$\boldsymbol{s}_{\text{unit}} = \Big(K_1^{\text{eff}} \boldsymbol{w}_1\Big) \,\|\, \Big(K_2^{\text{eff}} \boldsymbol{w}_2\Big) \,\|\, \cdots, \tag{62}$$

$$\boldsymbol{\theta} = \boldsymbol{\theta}_1 \,\|\, \boldsymbol{\theta}_2 \,\|\, \cdots, \tag{63}$$

$$\boldsymbol{C}_{\text{comb}} = \text{blockdiag}(\boldsymbol{C}_1, \boldsymbol{C}_2, \ldots), \qquad L_{\text{comb}} = \text{blockdiag}(L_1, L_2, \ldots), \tag{64}$$

so that the total expected mean is

$$\boldsymbol{\lambda}(\varepsilon^2, \boldsymbol{\theta}) = \boldsymbol{b} + L_{\text{comb}}\boldsymbol{\theta} + \varepsilon^2 \boldsymbol{s}_{\text{unit}}, \qquad \boldsymbol{C}_{\text{comb}} = L_{\text{comb}} L_{\text{comb}}^\top. \tag{65}$$

The bounded upper-limit statistic for the shared coupling, $\tilde{q}_{\varepsilon^2}(m)$, is defined exactly as in Eq. (45) after replacing the single-dataset amplitude with the common parameter of interest ε^2 and the single-dataset likelihood with $\mathcal{L}_{\text{comb}}$. The combination shares only the parameter of interest ε^2. The nuisance vectors $\boldsymbol{\theta}_d$ are independent between datasets, and no common radiative-fraction, luminosity, or resolution nuisance parameters are profiled across runs. We combine the datasets at the likelihood level rather than averaging upper-limit curves. At fixed ε^2,

$$-2\ln \lambda_{\text{comb}}(\varepsilon^2) = \sum_{d \in \mathcal{D}(m)} -2\ln \lambda_d(\varepsilon^2), \tag{66}$$

after profiling the independent nuisance blocks. The sensitivity gain therefore comes from the additive information content of statistically independent channels, each with its own detector response and normalization factor. This is the standard common-POI structure used in multi-channel HEP combinations [14, 19]. The combined upper limit is set directly on the shared coupling,

$$\varepsilon^2_{\mathrm{UL}}(m;\alpha) = \min\{\varepsilon^2 : \mathrm{CL}_s(\varepsilon^2) \leq \alpha\}, \tag{67}$$

with $\alpha = 0.10$ for the 90% CL result. The combined local significance is evaluated from the same profiled likelihood with the common signal vector $\varepsilon^2 \boldsymbol{s}_{\mathrm{unit}}$. Inverse-variance combinations are useful diagnostic proxies; they do not define the combined inference.

4.14 Global significance and the look-elsewhere treatment

The physics-facing local significance is the minimum one-sided p_0 in the scan,

$$p^{\mathrm{obs}}_{0,\min} = \min_m p^{\mathrm{obs}}_0(m), \qquad q^{\mathrm{obs}}_{0,\max} = \max_m q^{\mathrm{obs}}_0(m). \tag{68}$$

The cleanest global-significance definition is then scan based:

$$p^{\mathrm{toy}}_{\mathrm{global}} = \frac{1}{N_{\mathrm{toys}}} \sum_{t=1}^{N_{\mathrm{toys}}} \mathbf{1}\left[q^{(t)}_{0,\max} \geq q^{\mathrm{obs}}_{0,\max}\right], \tag{69}$$

with

$$Z^{\mathrm{toy}}_{\mathrm{global}} = \Phi^{-1}\left(1 - p^{\mathrm{toy}}_{\mathrm{global}}\right). \tag{70}$$

Until a dense full-scan calibration is available, we use an effective-trials approximation based on the scan span and detector resolution as an analytic look-elsewhere reference [20]:

$$p^{\mathrm{Sidak}}_{\mathrm{global}} = 1 - \left(1 - p^{\mathrm{obs}}_{0,\min}\right)^{N_{\mathrm{eff}}}, \qquad Z^{\mathrm{Sidak}}_{\mathrm{global}} = \Phi^{-1}\left(1 - p^{\mathrm{Sidak}}_{\mathrm{global}}\right). \tag{71}$$

For small $p^{\mathrm{obs}}_{0,\min}$, Eq. (71) reduces to the familiar $p_{\mathrm{global}} \simeq N_{\mathrm{eff}} p_{0,\min}$ form.

For the present result, Eq. (71) supplies a separate analytic look-elsewhere reference for the local scan. The effective number of trials is evaluated from the stated scan range and the mass-dependent resolution, so the approximation accounts for the strong correlation between adjacent 1 MeV-spaced hypotheses rather than treating every scan point as independent. It can be compared with the local p-value scan, but it is not a toy-calibrated global discovery probability. A background-only ensemble of the scan maximum, $q_{0,\max}$, is required for a discovery-level global claim; when that ensemble is available, Eq. (69) supersedes the analytic approximation.

4.15 Signal absorption and alternative background-profile treatments

We describe blind-window background uncertainty with additive Gaussian-constrained nuisance parameters derived from the GP mean and covariance. Normalizing the signal over its full line shape removes the bias from blind-window renormalization, but GP refitting or the nuisance model can still absorb part of an injected signal.

The most likely residual sources are the refitting freedom of the GP hyperparameters, especially the effective length scale, and the flexibility of the Gaussian-profile background nuisance itself. We therefore retain fixed-background and alternative Poisson treatments as targeted robustness tests if matched pseudoexperiments reveal unacceptable absorption.

4.16 Dimuon-threshold interpretation

The yield-to-coupling conversion in Eq. (38) is an electron-channel normalization: the fitted resonance amplitude is the $A' \to e^+e^-$ yield visible in the prompt invariant-mass spectrum. For the minimal kinetically mixed dark photon, this is also the full visible branching interpretation below the dimuon threshold, $m_{A'} < 2m_\mu = 211.3167\,\mathrm{MeV}$, where the $\mu^+\mu^-$ final state is kinematically closed. This is the same fixed-target convention used in the published HPS prompt searches [16, 5, 2]. Above threshold, the same electron-channel yield corresponds to a weaker minimal-model limit on ε^2 because the total width includes the newly open $A' \to \mu^+\mu^-$ mode [21, 22].

Neglecting the electron mass and remaining below the two-pion hadronic threshold, the charged-lepton width ratio is

$$R_{\mu/e}(m_{A'}) \equiv \frac{\Gamma(A' \to \mu^+\mu^-)}{\Gamma(A' \to e^+e^-)} = \sqrt{1 - \frac{4m_\mu^2}{m_{A'}^2}} \left(1 + \frac{2m_\mu^2}{m_{A'}^2}\right), \qquad m_{A'} > 2m_\mu, \tag{72}$$

with $R_{\mu/e} = 0$ below threshold. The inverse electron branching fraction used for the minimal-model reinterpretation is therefore

$$N_{\mathrm{eff}}^{\mathrm{BR}}(m_{A'}) \equiv \frac{1}{\mathcal{B}(A' \to e^+e^-)} = 1 + R_{\mu/e}(m_{A'}), \tag{73}$$

and an electron-channel coupling limit is mapped to the minimal visible-dark-photon convention by

$$\varepsilon_{\mathrm{min}}^2(m_{A'}) = N_{\mathrm{eff}}^{\mathrm{BR}}(m_{A'})\,\varepsilon_{ee}^2(m_{A'}). \tag{74}$$

This operation does not change the local fit, the signal-yield upper limit, the $\mathrm{CL_s}$ value, or the local p_0 scan; it changes only the coupling interpretation of an electron-channel result. Figure 21 shows the factor applied in the 90% CL result obtained with $\mathrm{CL_s}$. The factor is unity below $2m_\mu$, rises continuously above threshold, and reaches $N_{\mathrm{eff}}^{\mathrm{BR}} = 1.725$ at $250\,\mathrm{MeV}$ ($\mathcal{B}_{ee} = 0.580$).

The scan range used for a look-elsewhere statement must match the range of the quoted physics claim [20]. If the minimal-dark-photon exclusion is quoted only below 210 MeV, the final scan-level toy or effective-trials calibration should use that same below-threshold range. If the high-mass electron-channel scan is promoted to a 170 MeV to 250 MeV minimal-model result, the corresponding look-elsewhere calibration should include the full extended range as well.

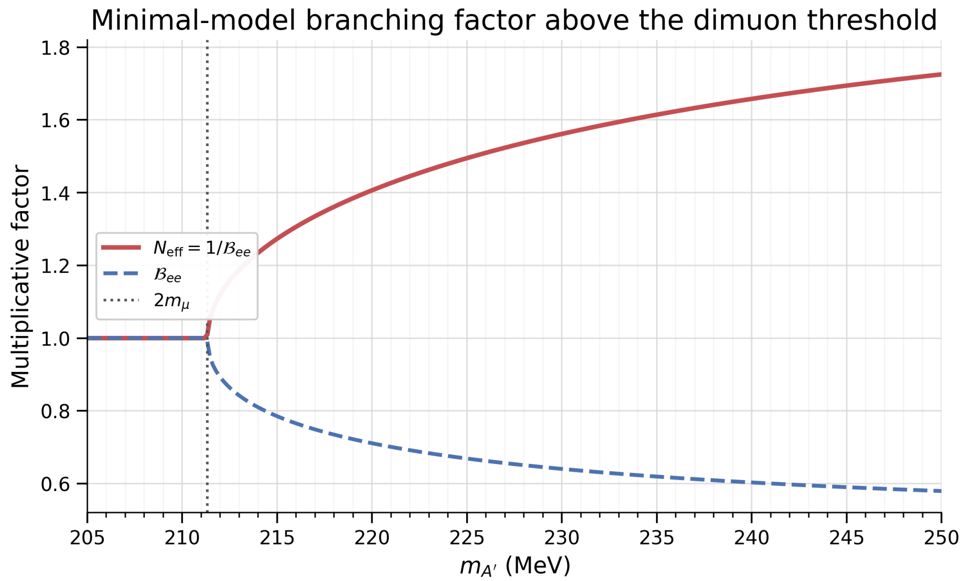


Figure 21: Minimal-model branching factor used to reinterpret electron-channel ε^2 limits above the dimuon threshold. The red curve shows $N_{\mathrm{eff}}^{\mathrm{BR}} = 1/\mathcal{B}(A' \to e^+e^-)$ and the blue dashed curve shows $\mathcal{B}(A' \to e^+e^-)$.

With the observable, background prediction, signal model, and likelihood specified, the remaining question is empirical: does this procedure create a false narrow signal, misstate its uncertainty, or absorb a signal that is actually present? We answer that question with pseudoexperiments in the next section.

5 Background-model validation with full-refit pseudoexperiments

5.1 Validation question and scope

If a smooth, data-like spectrum contains no resonance, does the GPR sideband fit return a spurious signal? If a known signal is added, does the same procedure recover it with an uncertainty of the right scale? For each pseudoexperiment we Poisson-fluctuate a smooth function fitted independently of the GP, refit the GP, and extract the signal. These tests exercise the count-space prediction, the mass-dependent blind and training masks, the full-range Gaussian signal template, the matched-reference injection convention, and a fresh GP optimization for every pseudo-spectrum. The statistical definitions are given in Section 4.11.

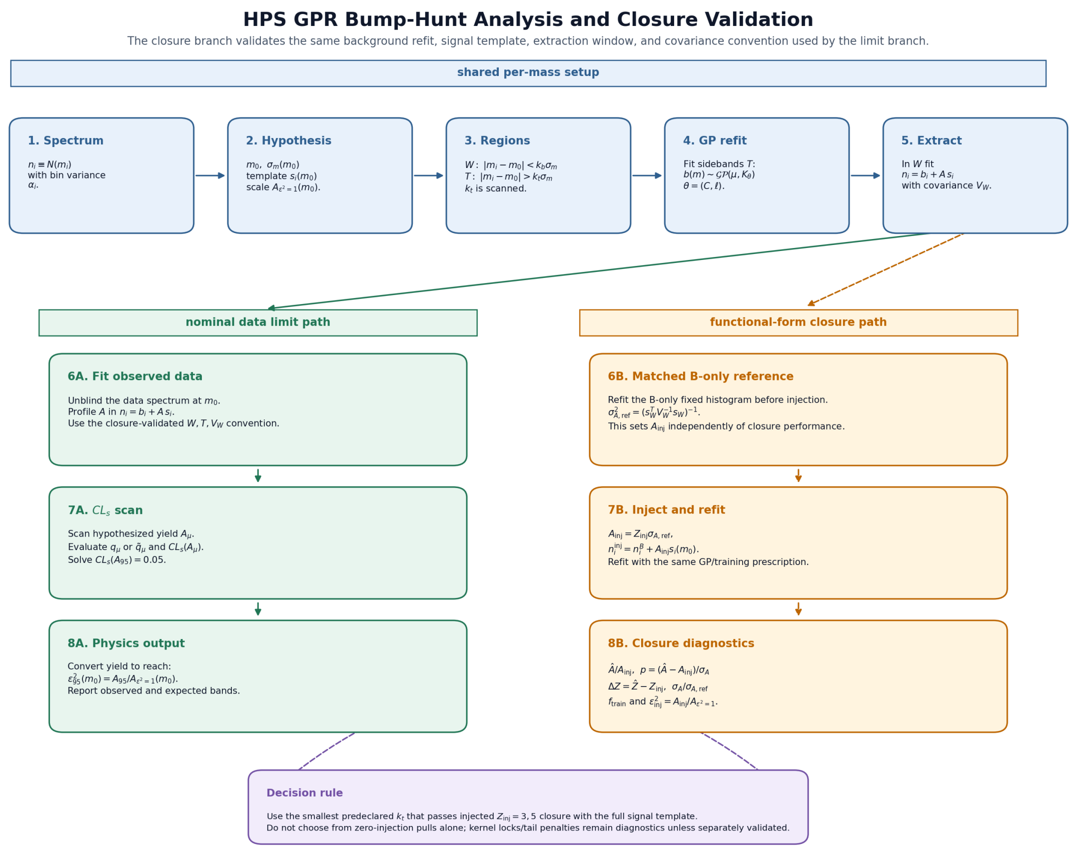


Figure 22: Analysis and validation flow. In the data path, sidebands around each mass hypothesis train the GP and the blinded bins enter the signal-plus-background likelihood. In the validation path, a smooth source-fitted mean is Poisson fluctuated, a known signal may be added, and the complete GP fit and extraction are repeated. The zero-injection ensemble tests spurious signal; nonzero injections test signal recovery and uncertainty calibration. These are extraction-closure tests, not confidence-limit coverage or a scan-wide significance calibration.

Each generating function is a smooth stress model fitted to an HPS source spectrum, not a unique physical parameterization of the trident continuum. The resulting closure statement is conditional on that model and the stated extraction settings; it does not measure bias in the observed data. We use four exposure ensembles. The 1% × 10 and 1% × 100 ensembles rescale the expected counts from the native 1% source by factors of ten and one hundred. The native-10% ensemble comes from a separately selected 10% sample, and 10% × 10 rescales its expected counts by a factor of ten. Because the two source samples also differ in selection and shape, equal effective exposure does not make them interchangeable.

5.2 Training exclusion and blind-window choice

The GP must not be trained on the core or near tails of the signal it is later asked to test. A Gaussian signal retains non-negligible weight outside a $1.64\sigma_m$ central interval, so training immediately beside that interval can offer signal-like counts to the background fit and reduce the recovered amplitude. The training-exclusion study therefore varied the sideband boundary before any observed limit was used to choose it.

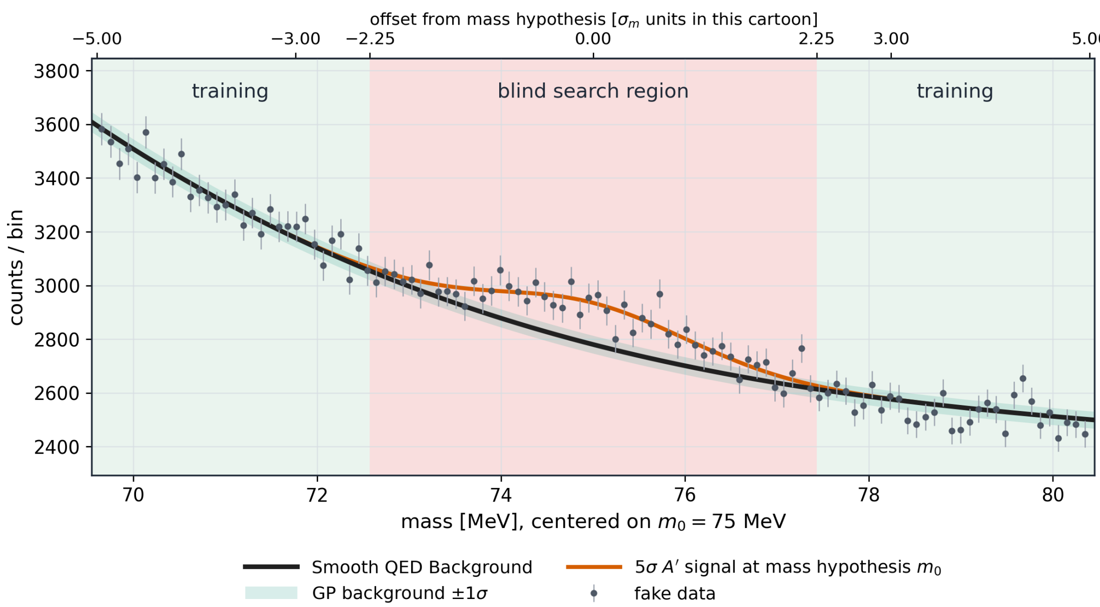


Figure 23: Selected $2.25\sigma_m$ blind and training-exclusion geometry. The GP is trained outside the widened interval, while all bins inside it enter the local signal-plus-background likelihood. This retains the signal tails for extraction without allowing those same bins to influence the sideband fit.

k_{train}	median expected ε^2_{95} proxy	$\langle \widehat{A}/A_{\text{inj}} \rangle_{z=3}$	$\langle \widehat{A}/A_{\text{inj}} \rangle_{z=5}$	max f_{train}
1.96	7.92×10^{-5}	0.730	0.715	0.0530
2.25	9.53×10^{-5}	0.961	0.943	0.0258
2.58	1.08×10^{-4}	1.052	1.036	0.0106
3.00	1.19×10^{-4}	1.049	1.037	0.00288

Table 5: Evidence used to choose the training exclusion. The train-2.25 row substantially reduces the fraction of the signal template available to the GP while recovering the injected amplitude and retaining useful local precision. The first column is a Gaussian-equivalent comparison proxy, not the final CL_s limit.

The widened $2.25\sigma_m$ interval is used for both the training exclusion and the extraction likelihood. Compared with a $1.64\sigma_m$ extraction plus a separate 1.64–$2.25\sigma_m$ guard band, this construction gives comparable closure while retaining the near signal tails in the likelihood. This gives one mass-dependent geometry for the background and signal fits.

5.3 Generating spectra and the role of functional forms

We generate the background pseudoexperiments independently of the GP used for their extraction. A positive analytic intensity is fitted to a native HPS mass spectrum, normalized on the specified toy support, integrated over every histogram bin, and Poisson fluctuated. A complete signal template is then added when $z > 0$. Every pseudo-spectrum is subsequently analyzed with a new GP sideband fit. This analytic truth/full-refit construction is the central background-model test because success is not guaranteed by generating toys from the GP model that will later fit them.

The 2015 and 2016 studies used independently fitted shifted sigmoid–power–exponential families. For the native 2021 1% source we used an unshifted sigmoid–power–exponential model. The four-exposure study used a generalized-gamma threshold model fitted independently to the native 1% and native 10% sources. These choices span smooth threshold, power-law, and broad-tail curvature without adding resonance-scale structure. Figure 24 shows the source spectra, selected means, and Poisson-ensemble construction.

The mass-resolved injection scans selected the lower RBF length-scale factors without using the observed limit curve. For the resulting settings, the aggregate zero-signal mean pulls are $+0.029$, -0.016, and -0.005 for the 2015, 2016 10%, and 2021 1% studies, respectively; the corresponding nonzero-injection signed averages are -0.223, -0.262, and -0.034. Mass-resolved mean, width, significance-residual, and recovery checks support these summaries. They motivate the chosen lower factors but do not replace direct mass-by-mass closure.

5.4 2021 validation ensembles at four effective exposures

The 2021 study uses 100 pseudoexperiments in each of four exposure ensembles and five mass hypotheses. The baseline extraction uses 40–300 MeV GP support, 50–250 MeV search range, $k_{\min} = 1.1$, $k_{\max} = 15$, five-bin local rebinning, log-mass/log-count preprocessing, the $2.25\sigma_m$ blind and training exclusions, two required sidebands, and twelve randomized optimizer restarts in addition to the nominal start. The four baseline ensembles use the same generalized-gamma threshold family, fitted independently to the native 1% and native 10% sources. Their source Pearson χ^2/ndf values are 1.108 and 2.781, respectively. The latter misdescription is largest through the steep threshold turn-on. We therefore use a threshold-refined model at 65 MeV while leaving the other mass points unchanged.

5.5 Targeted validation of the rising threshold

$1\% \times 10$ **threshold-refined model.** The first 65 MeV test uses a source-fitted near-threshold model. Over 35–80 MeV it is a logistic turn-on multiplied by an exponentiated degree-five Chebyshev expansion and is joined smoothly to the specified high-mass tail. The degree is selected before any GP extraction: the native-bin deviance per degree of freedom is 0.994 for the native 1% source and the five-bin value is 0.850, with no parameter-bound contact. Its disclosed 80–300 MeV tail Poisson deviance per degree of freedom is 1.122, so this selected 1%-source truth is broadly data-like over the complete 40–300 MeV support rather than only in the fitted turn-on interval. An initial 20-background development subset gave a mean pull of 0.300 and a width of 1.109. We

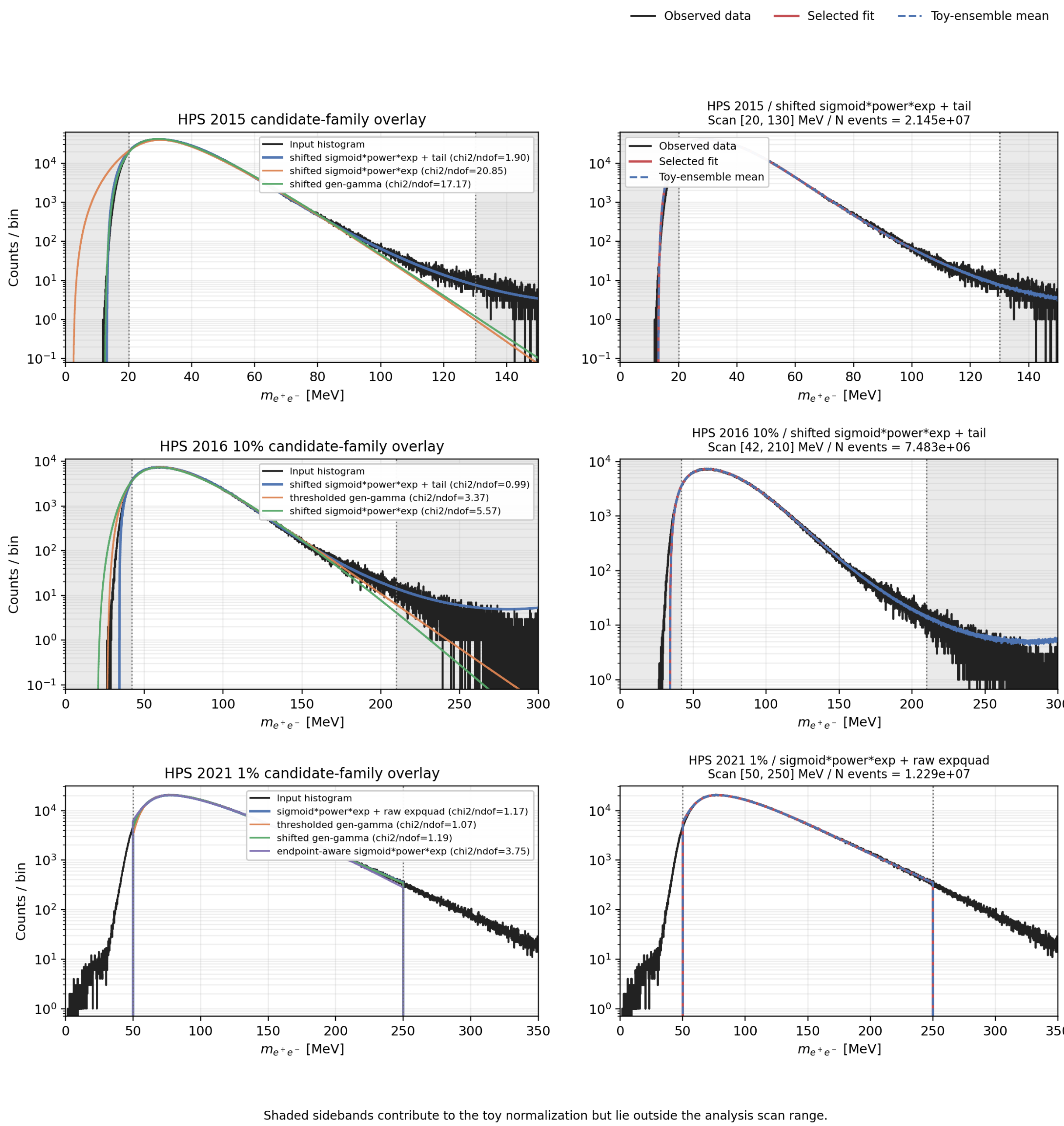


Figure 24: Analytic generating means used in the year-matched functional-form studies. Each dataset row compares the source spectrum with the candidate families and then shows the selected mean against the generated toy ensemble. The functions define smooth positive pseudo-data means and are not asserted to be unique physical models of the continuum.

Displayed ensemble	Generating model	GP support	Role in the test
1% × 10, except 65 MeV	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
1% × 10, 65 MeV	degree-five threshold-turn-on continuation	40–300 MeV	threshold-refined 65 MeV test
1% × 100, all masses	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
native 10%, except 65 MeV	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
native 10%, 65 MeV	threshold-focused sigmoid–power–exponential mean	30–300 MeV	extended-support 65 MeV test of the measured turn-on
10% × 10, all masses	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test

Table 6: Models used in the 2021 background-validation figures. At 65 MeV, the two threshold-focused tests replace the baseline points rather than being pooled with them. Distinct markers identify the change of generating model and, for native 10%, the extension of the GP support.

then analyzed an independently specified continuation, bringing the total to 100 backgrounds and allowing 90% intervals to be reported.

Native-10% extended-support model. The second 65 MeV test models the measured 30–40 MeV shoulder and analyzes it with the complete 30–300 MeV GP support. Its threshold component is a logistic turn-on times a degree-six exponentiated Chebyshev form fitted over 30–80 MeV, joined with a C^2 taper to a smooth high-mass continuation. In the native 10% source, the local native-bin and five-bin deviances per degree of freedom are 1.140 and 1.421. Thus the generating spectrum is data-like in the threshold-sensitive region even though the baseline family is not. Its 85–300 MeV tail deviance per degree of freedom is 6.220; this native-10% ensemble therefore relies on the weak-correlation and tail-influence qualification below rather than on a support-wide goodness-of-fit claim.

This study tests the 65 MeV extraction with 30–300 MeV support; it is separate from the lower-edge study that selected 36–300 MeV for the 2021-only analysis. The question here is whether a data-like generating mean that passes the local source-fit tests can be processed by the complete extraction without material signal bias. With log preprocessing, the direct RBF correlation between masses m and m' is

$$\rho(m, m') = \exp\left[-\frac{\log^2(m'/m)}{2\ell^2}\right]. \tag{75}$$

At 65 MeV, the optimized length scales in the two targeted ensembles span 0.2784–0.3185 for 1% × 10 and 0.2468–0.2691 for native 10%, with medians 0.2963 and 0.2592. The corresponding direct correlations with 150 MeV span 1.10–3.19% and 0.321–0.800%, respectively. Across the complete 55–70 MeV set the largest direct correlations with 150 MeV occur at 70 MeV and are 5.71% and 1.81%. The correlation decreases rapidly above 150 MeV. These results quantify why a threshold-focused source fit can provide a meaningful local test while the generated spectrum and extraction still retain the full GP support.

The correlation calculation is a fixed-hyperparameter statement, not a claim that high-mass bins have mathematically zero influence. All retained bins still enter the global marginal likelihood and can shift the optimized length scale or kernel constant. A same-input diagnostic makes the aggregate effect explicit: after fixing the selected hyperparameters, removing the training bins above 150 MeV shifts the 65 MeV pull by a mean of -0.130 with a paired 90% interval $[-0.151, -0.110]$ in $1\% \times 10$, and by -0.195 $[-0.211, -0.179]$ in native 10%. The maximum absolute shifts are 0.376 and 0.400. In the first ensemble the mean changes from 0.139 to 0.009; in the second it changes from -0.246 to -0.441, whose 90% interval $[-0.611, -0.271]$ crosses the descriptive $|\overline{p}| = 0.5$ boundary used below. Counts above 150 MeV therefore have modest, statistically resolved collective influence even though every pairwise RBF correlation is small. This diagnostic holds the chosen hyperparameters fixed and does not test their global-likelihood response to tail removal.

The optimized kernels explain why a threshold-focused source fit is informative for the full extraction: the low-mass result is directly coupled most strongly to the turn-on region, while the generated spectrum, specified smooth continuation, and full extraction retain the actual support. The restricted source fit thus provides a targeted conditional validation of local 65 MeV behavior with the stated support. It does not establish tail-independence or universal GP closure for every possible high-mass truth.

5.6 Matched refits and numerical selection

For every background, mass, and generating model, an independently optimized background-only fit supplies the matched reference uncertainty. Signals with $z = 1, 3, 5$ are generated with mean amplitude $A_{\text{inj}} = z\sigma_{A,\text{ref}}$, their full Gaussian templates are added to the same background counts, and the GP is refitted after applying the same training mask. The injected amplitude is therefore calibrated to the local background-only fit, whereas the pull denominator is always the uncertainty returned by the injected fit.

Whether a fit is kept never depends on its fitted signal or pull. Repeated attempts use the same pseudo-data with independent optimizer starts. We choose among them using the GP log marginal likelihood, reproducibility of the length scale and kernel constant, covariance validity, uncertainty-ratio checks, and contact with the kernel bounds. The choice never uses $\widehat{A}$, the pull, agreement with the injected value, ε^2, or a local p value. A statistical outlier is retained whenever it passes these numerical checks.

The baseline study retained 7994 of 8000 fits; every ensemble contains 99 or 100 usable fits. The six rejected fits failed reproducibility checks. Each threshold-focused study contains 400 usable fits—100 backgrounds at each of $z = 0, 1, 3, 5$—with no rejection and no kernel-bound contact. Covariance, count, and pull checks agree, and the numerical selection is made without reference to the extracted physics result.

5.7 Closure observables and decision rules

At zero injection, the signed pull $p = \widehat{A}/\widehat{\sigma}_A$ is a spurious-signal statistic. The ensemble mean tests whether the extraction is centered and the sample standard deviation tests whether the returned uncertainty has the appropriate scale. At nonzero injection, the pull of Eq. (51) tests centering about the known amplitude and the paired response of Eq. (52) separates recovery of the added signal from a background-specific zero-signal offset. Strengths $z = 0, 1, 3, 5$ reuse the same background pseudo-spectrum and are consequently correlated; they are one response curve, not four independent confirmations.

For each 100-background ensemble at one mass and signal strength, we report the usable count

n, the pull mean and its two-sided 90% Student-t interval, and the pull width with its exact normal-theory 90% chi-square interval. An ensemble is reported only if at least 95 of its 100 fits meet the numerical criteria. We use a Holm correction for the two planned threshold-focused points at $z = 0$, and a separate correction for the twenty displayed four-exposure, five-mass points. The correlated nonzero injections are response diagnostics and do not enter either family. We flag a mean only when its Holm-adjusted two-sided p value is below 0.05 and $|\overline{p}| \geq 0.2$.

The practical centering tolerance $|\overline{p}| < 0.5$ was chosen after the 100-background results were available. It is a descriptive criterion, not a predeclared equivalence test. We report it alongside the t/Holm results, which are more sensitive to small but statistically resolved offsets. For the two targeted points we also ask whether the full 90% interval on the mean lies inside the practical band. A pull width is consistent with one only when its 90% interval contains one. None of these tests measures $\mathrm{CL_s}$ coverage.

5.8 Spurious-signal results from 100 pseudoexperiments

Table 7 gives the two targeted 65 MeV results. The baseline rows show why the threshold-focused tests were needed; the two sets of pulls are not combined. Both targeted samples exceed the 95-of-100 reporting threshold and pass the numerical checks above. We compare the results under the original statistical test, the post-result practical $|\overline{p}| < 0.5$ criterion, and the separate unit-width diagnostic.

65 MeV ensemble and model	n	$\overline{p}$ [90% CI]	s_p [90% CI]	centering
1% × 10, baseline	100	0.7890 [0.6293,0.9488]	0.9622 [0.8625,1.0907]	outside band
1% × 10, threshold-refined, full sample	100	0.1394 [−0.0499,0.3288]	1.1404 [1.0222,1.2927]	inside band
1% × 10, threshold-refined, independent continuation	80	0.0993 [−0.1150,0.3135]	1.1514 [1.0196,1.3265]	inside band
native 10%, baseline	100	0.7160 [0.5452,0.8867]	1.0284 [0.9218,1.1657]	outside band
native 10%, extended support, full sample	100	−0.2463 [−0.4165,−0.0761]	1.0253 [0.9190,1.1622]	inside band†
native 10%, extended support, independent continuation	75	−0.2840 [−0.4943,−0.0737]	1.0934 [0.9646,1.2661]	inside band

Table 7: Background-only closure at 65 MeV. Intervals are two-sided 90% intervals: Student-t for the mean and exact normal-theory chi-square for the population standard deviation. The two full-sample targeted intervals lie inside the post-result $[-0.5, 0.5]$ practical band, whereas the baseline means do not. †The native-10% full sample nevertheless has targeted Holm-2 $p = 0.03632$ and $|\overline{p}| > 0.2$; statistical resolution of a small offset and practical centering are distinct statements. The 1% × 10 width interval excludes one in both the full sample and independent continuation.

Both threshold-focused samples meet the post-result practical centering criterion in the full sample and in the independent continuation. The 1% × 10 pull width is mildly high, while the native-10% mean retains a statistically resolved negative offset in the targeted two-point test. We therefore report practical centering, evidence for a nonzero offset, and pull-width calibration as separate results.

Across the full display, all twenty zero-signal mean pulls lie inside $[-0.5, 0.5]$; the largest magnitude is 0.338 for 10% $\times$ 10 at 180 MeV. That point is the only flag after the separate twenty-point Holm correction. With the correlated nonzero injections included, all eighty mean pulls remain inside the practical band and 75 of their 90% intervals are wholly contained. Exact unit-width calibration is less uniform: three of twenty zero-signal width intervals and fifteen of eighty all-strength intervals exclude one.

Figure 26 directly compares the baseline and targeted 65 MeV ensembles, showing why the threshold-specific study was needed.

5.9 Recovery of injected signals

A centered zero-signal ensemble is necessary but not sufficient: a background model could remove the false offset and still absorb a real narrow signal. The nonzero rows therefore repeat the full GP fit after adding matched one-, three-, and five-reference-sigma signals. Figure 27 displays the mean and width at every strength and mass; Table 8 gives the numerical 65 MeV replacement results, including the paired response $R = (\widehat{A}_z - \widehat{A}_0)/A_{\mathrm{inj}}$.

The paired response separates signal recovery from the additive background offset. In the two threshold-focused tests, the mean response is stable at 0.966–0.969 over $z = 1, 3, 5$, with mutually consistent 90% intervals and no worsening with strength. The full procedure therefore recovers 96.6–96.9% of the tested amplitude, corresponding to a stable 3.1–3.4% attenuation. We retain that attenuation in the interpretation. Agreement across strengths is not counted multiple times because the same 100 background realizations are reused.

65 MeV model	z	$\overline{p}$ [90% CI]	s_p [90% CI]	$\overline{R}$ [90% CI]
1% $\times$ 10, threshold turn-on	0	0.139 [−0.050,0.329]	1.140 [1.022,1.293]	—
	1	0.107 [−0.082,0.297]	1.141 [1.023,1.293]	0.9678 [0.9643,0.9713]
	3	0.039 [−0.149,0.227]	1.134 [1.016,1.285]	0.9665 [0.9646,0.9684]
	5	−0.025 [−0.216,0.165]	1.147 [1.028,1.300]	0.9670 [0.9654,0.9686]
native 10%, 30–300 support	0	−0.246 [−0.417,−0.076]	1.025 [0.919,1.162]	—
	1	−0.280 [−0.450,−0.109]	1.025 [0.919,1.162]	0.9664 [0.9629,0.9700]
	3	−0.346 [−0.515,−0.177]	1.019 [0.913,1.155]	0.9666 [0.9649,0.9684]
	5	−0.402 [−0.571,−0.234]	1.015 [0.909,1.150]	0.9686 [0.9669,0.9703]

Table 8: Full-sample 65 MeV extraction at all matched-reference strengths for the two threshold-focused models. Mean intervals are two-sided 90% Student-t intervals and width intervals are exact normal-theory two-sided 90% chi-square intervals. The response summaries are paired background-by-background and use two-sided 90% Student-t intervals. All strengths share the same backgrounds and are correlated. The 1% $\times$ 10 widths are mildly high at every strength; the native-10% widths are compatible with one at this precision.

The validation above assumes fixed GP supports. For the 2021-only scan, we must also choose where that support begins.

5.10 Selecting the 2021 GP support

The signal search tests masses from 50 to 250 MeV, but the GP also needs data outside that interval to constrain the background near its endpoints. At low mass, the native 2021 spectrum has a sparse shoulder followed by a rapid turn-on: its 0.125 MeV bins contain only a few hundred events around

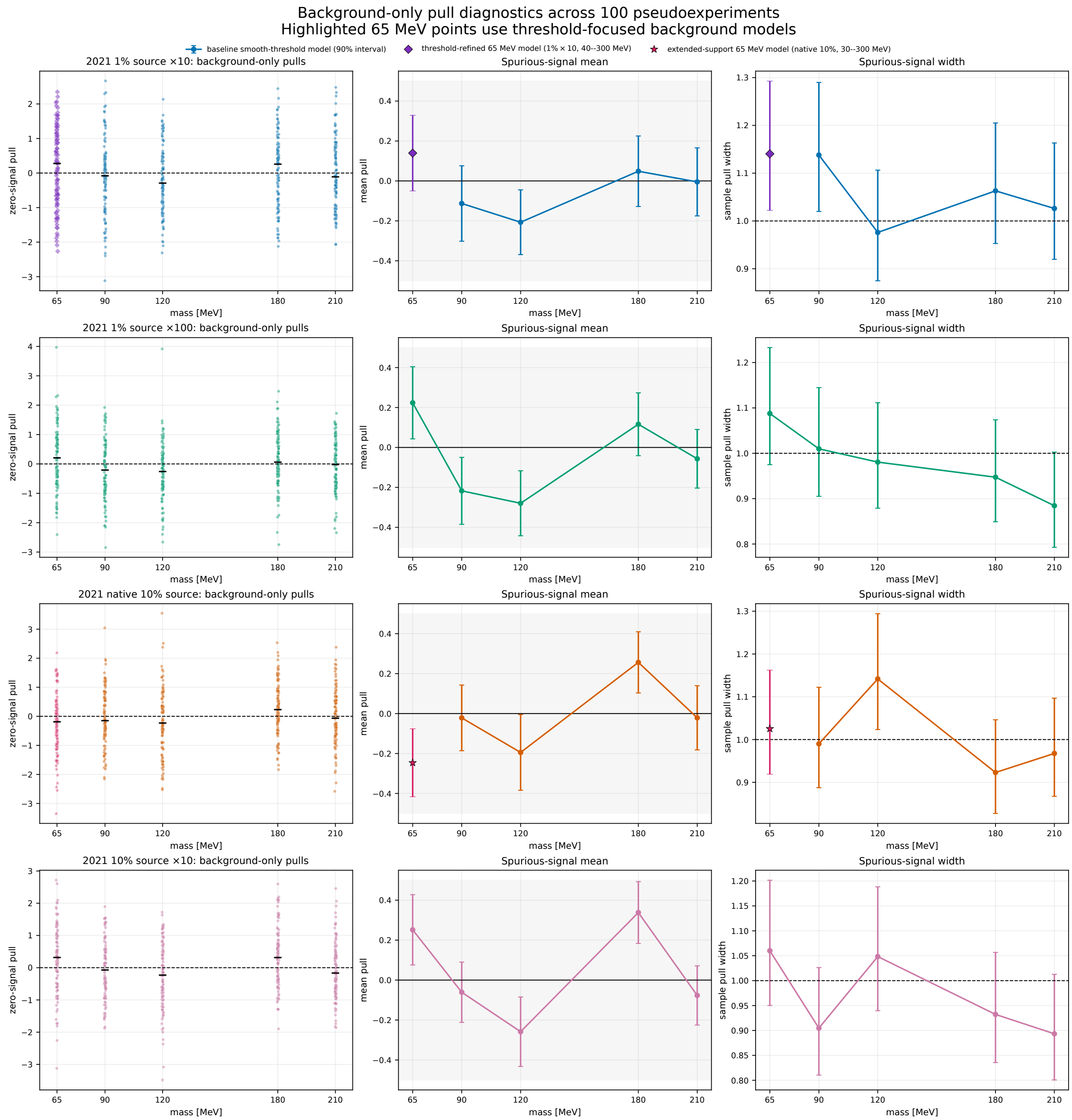


Figure 25: Zero-signal diagnostics for four 2021 exposure ensembles, with 100 pseudoexperiments per point. Circular markers show the baseline smooth-threshold model. At 65 MeV, diamonds and stars show the two threshold-focused tests defined in Table 6; they replace, rather than pool with, the baseline points. Bars are two-sided 90% intervals. The gray mean band is the post-result ±0.5 practical centering tolerance.

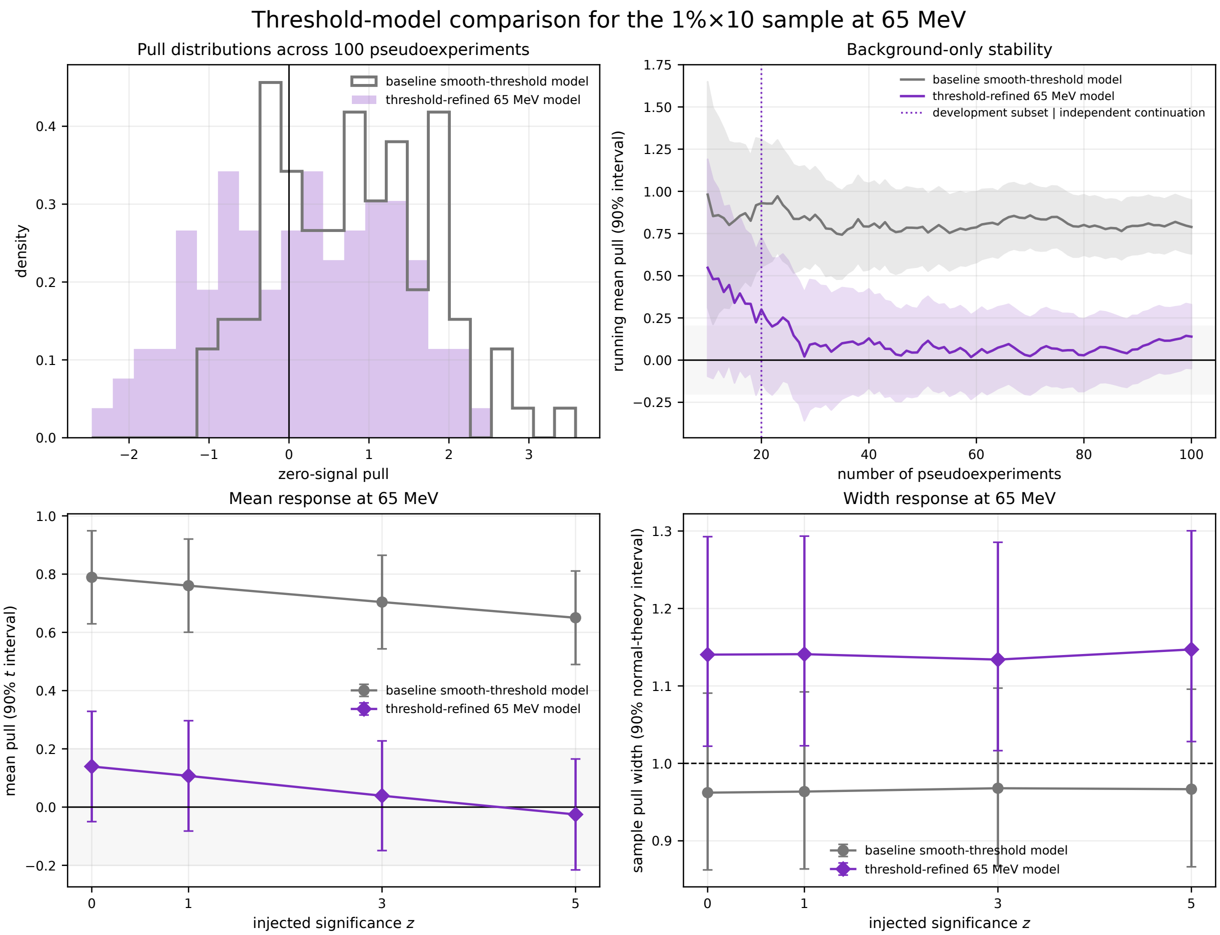


Figure 26: Targeted $1\% \times 10$ test at 65 MeV, comparing the threshold-refined and baseline smooth-threshold models. The upper panels show the zero-injection pull distribution and the running mean; the lower panels show the mean and width across the correlated $z = 0, 1, 3, 5$ extractions. Mean intervals are two-sided 90% Student-t intervals and width intervals are exact normal-theory 90% chi-square intervals. The full sample contains the 20-background development subset; the independent continuation in Table 7 gives the same qualitative result. The gray ± 0.2 band is a narrow visual reference, not the ± 0.5 practical criterion used in the text.

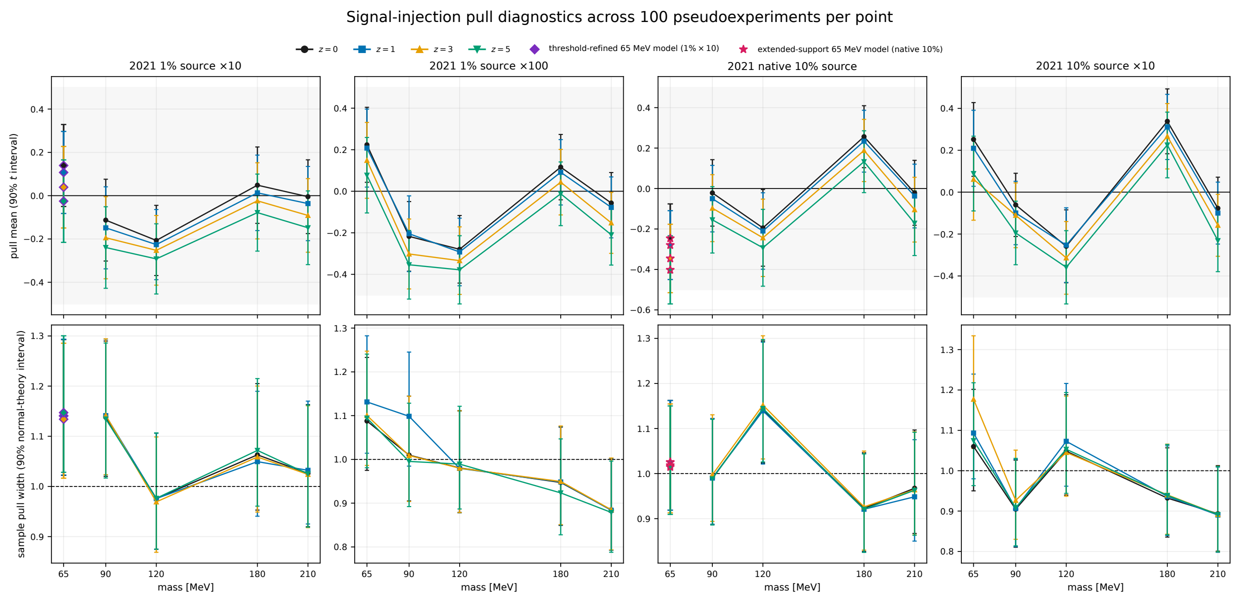


Figure 27: Background-validation summary. Columns correspond to $1\% \times 10$, $1\% \times 100$, native 10%, and $10\% \times 10$; the upper row shows pull means and the lower row pull widths as functions of mass. The four series are $z = 0, 1, 3, 5$, with zero injection visually emphasized and the correlated nonzero response series drawn more lightly. Circular markers show the baseline model; at 65 MeV, diamonds and stars show the targeted studies defined in Table 6. Mean bars are two-sided 90% Student-t intervals and width bars are exact normal-theory two-sided 90% chi-square intervals. The four strengths share backgrounds. The gray mean bands show the post-result ± 0.5 practical centering tolerance.

30–32 MeV and several thousand by 40 MeV. We therefore compared several lower support edges while holding the upper edge at 300 MeV and leaving the search interval unchanged. The aim was to exclude the least stable part of the turn-on without discarding useful sideband information.

5.10.1 Source shape and candidate supports

Figure 28 shows the source spectrum and the tested lower edges. The 30 MeV setting is retained as a control but was not eligible for selection; the candidate edges were 32, 34, 36, 38, and 40 MeV.

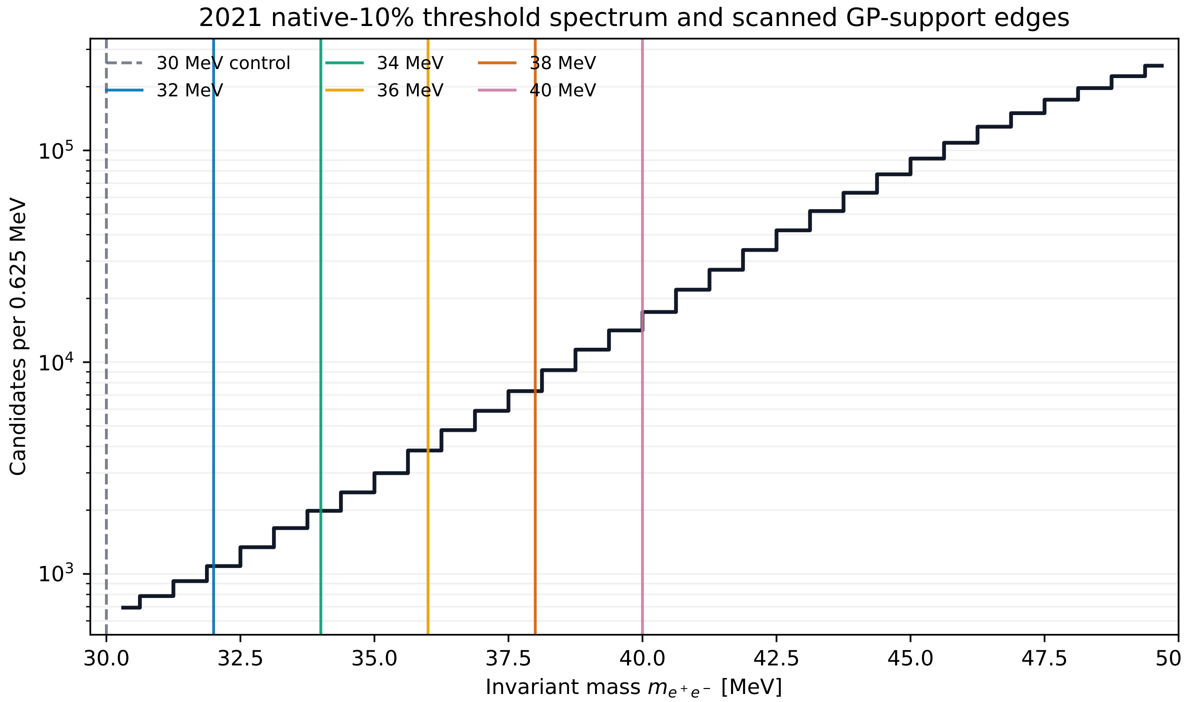


Figure 28: Native 2021 10% source histogram near threshold. Vertical lines mark the 30 MeV control and the five candidate lower edges of the GP support. The logarithmic vertical scale makes the low-count shoulder and rapid turn-on visible.

5.10.2 Initial scan

The first stage reuses 100 background-only pseudoexperiments generated from the fitted source spectrum in the four-exposure study; no new pseudo-data are generated. At each support edge, matched-reference extractions are performed at 55, 60, 65, and 70 MeV with 0-, 2-, and 5-σ_A injections. The scan changes only the lower support edge. Binning, preprocessing, blind and training masks, kernel bounds, signed extraction, and optimizer checks remain fixed as described in Sections 4 and 5. For repeated optimizer starts, we retain the fit using reproducibility, covariance validity, kernel diagnostics, and GP log marginal likelihood; neither the pull nor any limit or p-value enters that choice.

The first 25 backgrounds were processed at all six edges. All 1800 fits passed the numerical checks, with no invalid covariance matrix or kernel-bound contact. Under the original rule, an edge was rejected when any mass–injection combination had $|\overline{p}| \geq 0.5$ and its two-sided 90% Student-t

interval excluded zero. Every candidate was rejected. Table 9 records that outcome under the rule used for the initial scan.

Table 9: Initial 25-background minimax summary. Every candidate is rejected by the original injection criterion. The 30 MeV row is a control and was never eligible for selection.

Lower edge [MeV]	Worst $\lvert\overline{p}\rvert$	Worst zero-signal $\lvert\overline{p}\rvert$	Initial rule
30	1.624	1.300	fail (control)
32	1.530	1.331	fail
34	1.447	1.277	fail
36	1.134	0.959	fail
38	1.072	0.864	fail
40	1.152	1.152	fail

5.10.3 Practical criterion and independent confirmation

After the first stage, we adopted a less stringent practical criterion: at least 9 of the 12 mass–injection means and at least 3 of the 4 zero-signal means had to satisfy $\lvert\overline{p}\rvert < 0.75$, and no mean could reach 1.25 in magnitude. The numerical checks, the minimax score, and the 0.10 tie margin were left unchanged. Among candidates within that margin, the lower edge was preferred so that the fit retained more data. Because this criterion was formulated after the first 25 backgrounds had been examined, it is a post-selection practical rule, not a predeclared equivalence test or a coverage criterion. It was fixed before the remaining 75 backgrounds were processed.

Only the 36 and 38 MeV candidates pass the practical rule in the first stage. The 38 MeV edge has the smaller worst pull, but 36 MeV lies within the retained tie margin and is therefore selected for confirmation. The independent continuation uses backgrounds 25–99 at 34, 36, and 38 MeV. At 36 MeV, 10 of 12 means and all four zero-signal means satisfy the 0.75 condition in the continuation alone; the worst magnitude is 0.901. In the complete 100-background sample, the corresponding counts are 9 of 12 and 3 of 4, with a worst magnitude of 0.960. Every fit passes the numerical checks. With the criterion already fixed, we chose 36–300 MeV for the subsequent 2021-only analysis without further tuning.

At the selected edge, the full-sample means at 55 MeV are -0.773, -0.841, and -0.960 for the 0-, 2-, and 5-σ_A injections. Their 90% intervals exclude zero, so the low-mass under-recovery is retained as a real limitation of this test. Across the nine mass–injection combinations from 60 to 70 MeV, the means range from -0.110 to $+0.325$ and the pull widths from 0.939 to 1.034. The 65 MeV zero-signal mean is 0.253, with a 90% interval of $[0.086, 0.421]$.

Each numerically valid pseudoexperiment also provides a bounded Wald $\widetilde{q}_\mu$ 90% $\mathrm{CL_s}$ yield limit. We use those values only to check that the limit is stable under nearby support choices; the smallest limit never determines the selected edge. This study therefore supports the 36–300 MeV choice for the declared fitted-source ensemble, with the 55 MeV caveat above. It does not establish direct coverage or truth-independent background closure.

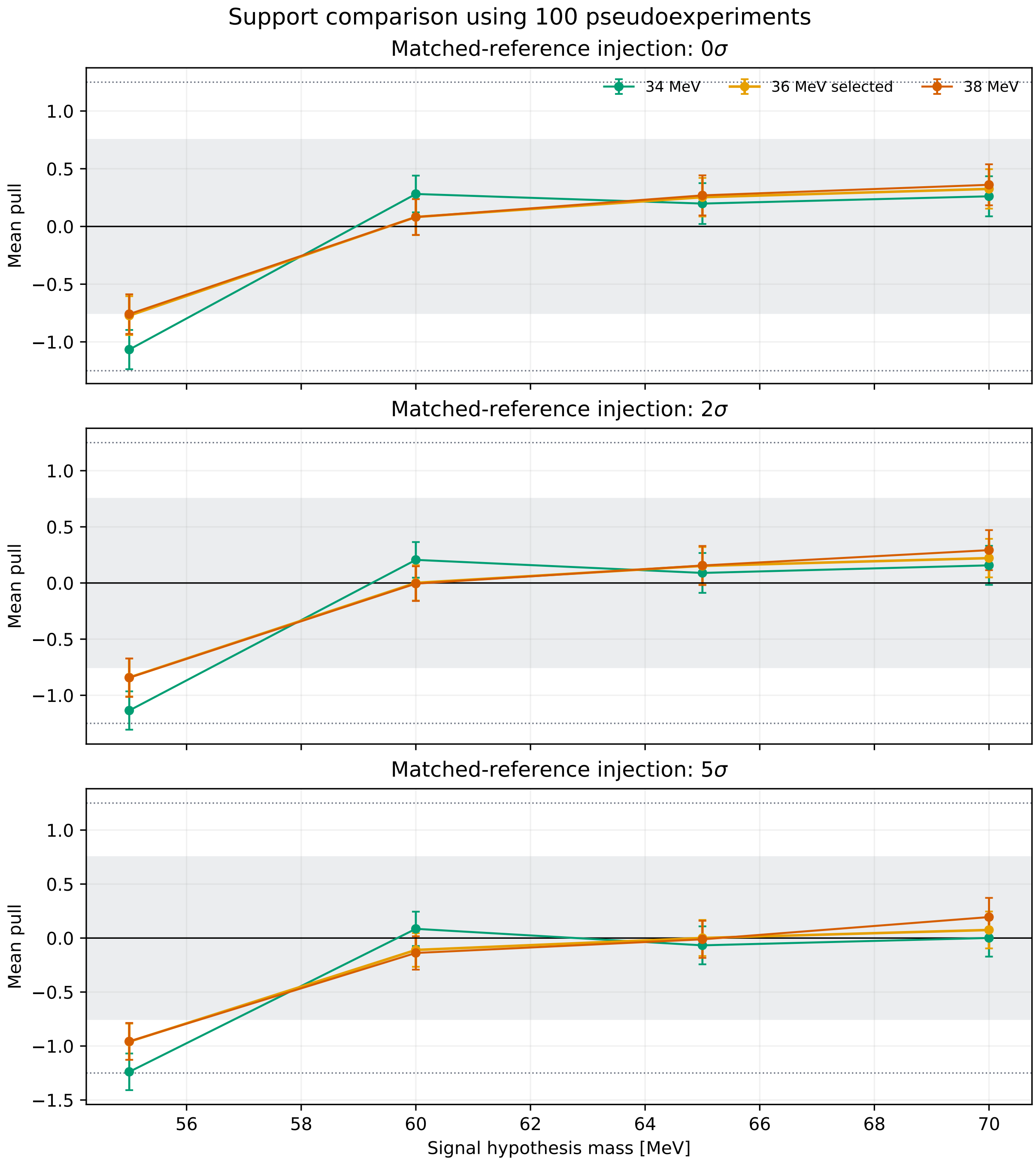


Figure 29: Mean pulls in the complete 100-background sample for the selected edge and its two neighboring candidates. Error bars are two-sided 90% Student-t intervals on the mean. The shaded band denotes the post-selection $|\overline{p}| < 0.75$ practical region; dotted lines show the magnitude-1.25 guard. The remaining exceptions are confined to the 55 MeV hypothesis.

5.11 What the validation establishes

For the source-like spectra tested here, the GP extraction is practically centered and recovers an injected narrow signal with a stable response of 96.6–96.9%. All twenty zero-signal mean pulls and all eighty correlated injection-strength mean pulls lie inside the post-result ± 0.5 band. The result rests on an independent generating model, a fresh sideband fit for every pseudoexperiment, and numerical choices made without reference to the fitted signal or pull.

The departures from ideal behavior are equally informative. The response is 3.1–3.4% low; three of twenty zero-signal width intervals and fifteen of eighty all-strength width intervals exclude one; and the native-10%, 65 MeV test retains a small resolved offset. The support study also finds under-recovery at 55 MeV. These tests therefore establish conditional extraction closure for the spectra and supports studied, not zero bias for arbitrary smooth backgrounds, exact pull calibration, confidence-limit coverage, bias in the observed data, or a scan-global significance calibration.

6 Selected observed results

The pseudoexperiments in Section 5 set the limits of the statistical interpretation. Within those limits, this section compares the observed scans. In the figures that follow, the top and middle panels show how many signal events remain compatible with the data at 90% $\mathrm{CL_s}$ and the corresponding upper limit on the kinetic-mixing parameter ε^2. The bottom panels show the local asymptotic background-only p_0. A small p_0 identifies an upward fluctuation at one tested mass. Only observed curves are shown. There are no expected-limit bands in any figure in this section.

Two configurations appear in these comparisons. The full-2015 and full-2016 results, and every multi-campaign combination, use the three-campaign analysis; its 2021 input is the native 10% sample with 40–300 MeV GP support. The separate 2021 10% curve uses the 2021-only analysis with the selected 36–300 MeV support. We do not substitute that later support into the combinations. The 2016 10% and 2021 1% curves show how the standalone scans behave at lower exposure; they do not enter the combinations.

6.1 Individual data sets

Figure 30 compares five standalone scans: full 2015, 2016 10%, full 2016, 2021 1%, and 2021 10%. The top panel reports the signal-yield limit before translating it into a coupling; the middle panel applies the campaign-specific luminosity, efficiency, resolution, radiative fraction, and branching inputs needed for the ε^2 limit. Keeping both panels visible separates what comes directly from the statistical fit from what also depends on the physics normalization.

The lowest three mass hypotheses in the 2021 1% scan, 50–52 MeV, lie immediately above its lower search boundary and are sensitive to the treatment of the GP support edge. They are therefore shown for completeness but not promoted as candidate-level evidence. This distinction matters especially in a multi-curve plot: a formally small local value at an analysis boundary can be visually striking even when it is known to be a modeling diagnostic.

After those three edge points are set aside, the smallest local value in the 2021 1% series occurs at 244 MeV, $p_0 = 2.616405 \times 10^{-3}$ ($Z = 2.79234$). In the 2021-only 10% series, the minimum occurs at 78 MeV, $p_0 = 2.474845 \times 10^{-3}$ ($Z = 2.81029$).

6.2 Shared-coupling combinations

The combinations in Figure 31 profile one common ε^2 while allowing each campaign to retain its own expected signal yield and background model. They are simultaneous likelihood results, not pointwise envelopes or arithmetic averages of the standalone limits. The three pairwise curves are full 2015 + 2021 10%, full 2016 + 2021 10%, and full 2015 + full 2016. The all-three curve is full 2015 + full 2016 + 2021 10%, evaluated where all three campaigns overlap. It does not imply a 2021 full-statistics result.

Using a common coupling is important physically. A candidate signal would have the same ε^2 in each campaign, but its expected event count would change with beam conditions, exposure, acceptance, and mass resolution. The simultaneous likelihood enforces that relationship while allowing each data set to contribute according to its own reach.

6.3 Local asymptotic scan

Figure 32 collects the local asymptotic p_0 series without the two limit panels. This view makes common mass-local structure easier to compare across data sets and combinations. Every point

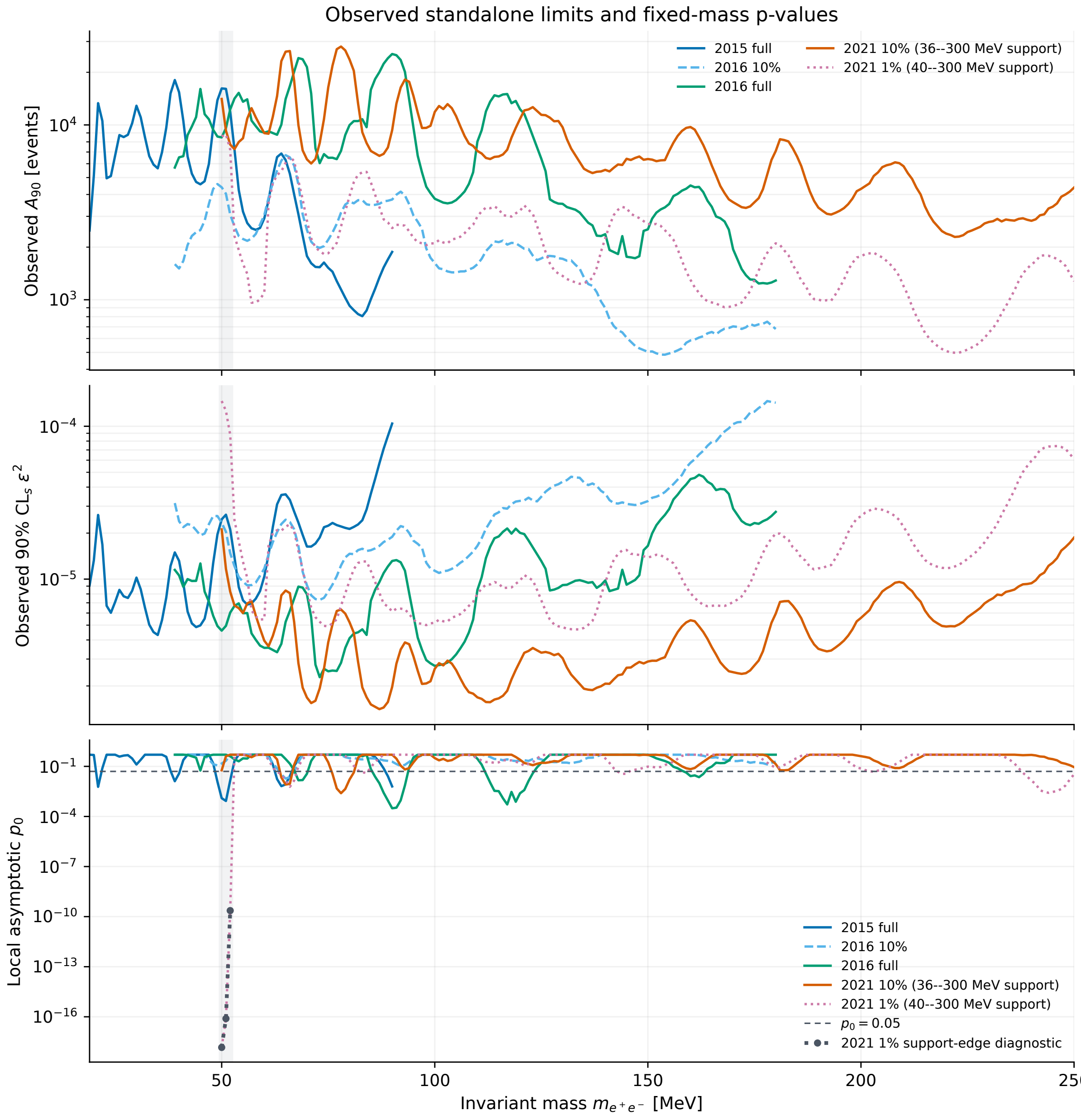


Figure 30: Standalone observed results for full 2015, 2016 10%, full 2016, 2021 1%, and 2021 10%. From top to bottom, the panels show the 90% asymptotic $\mathrm{CL_s}$ signal-yield upper limit, the corresponding electron-channel ε^2 upper limit, and the local asymptotic p_0. The full-2015 and full-2016 curves use the three-campaign configuration; the 2021 10% curve uses the 2021-only configuration. No expected-limit bands are shown. The marked 50–52 MeV portion of the 2021 1% scan is retained as a GP-support-edge diagnostic and is excluded from candidate interpretation.

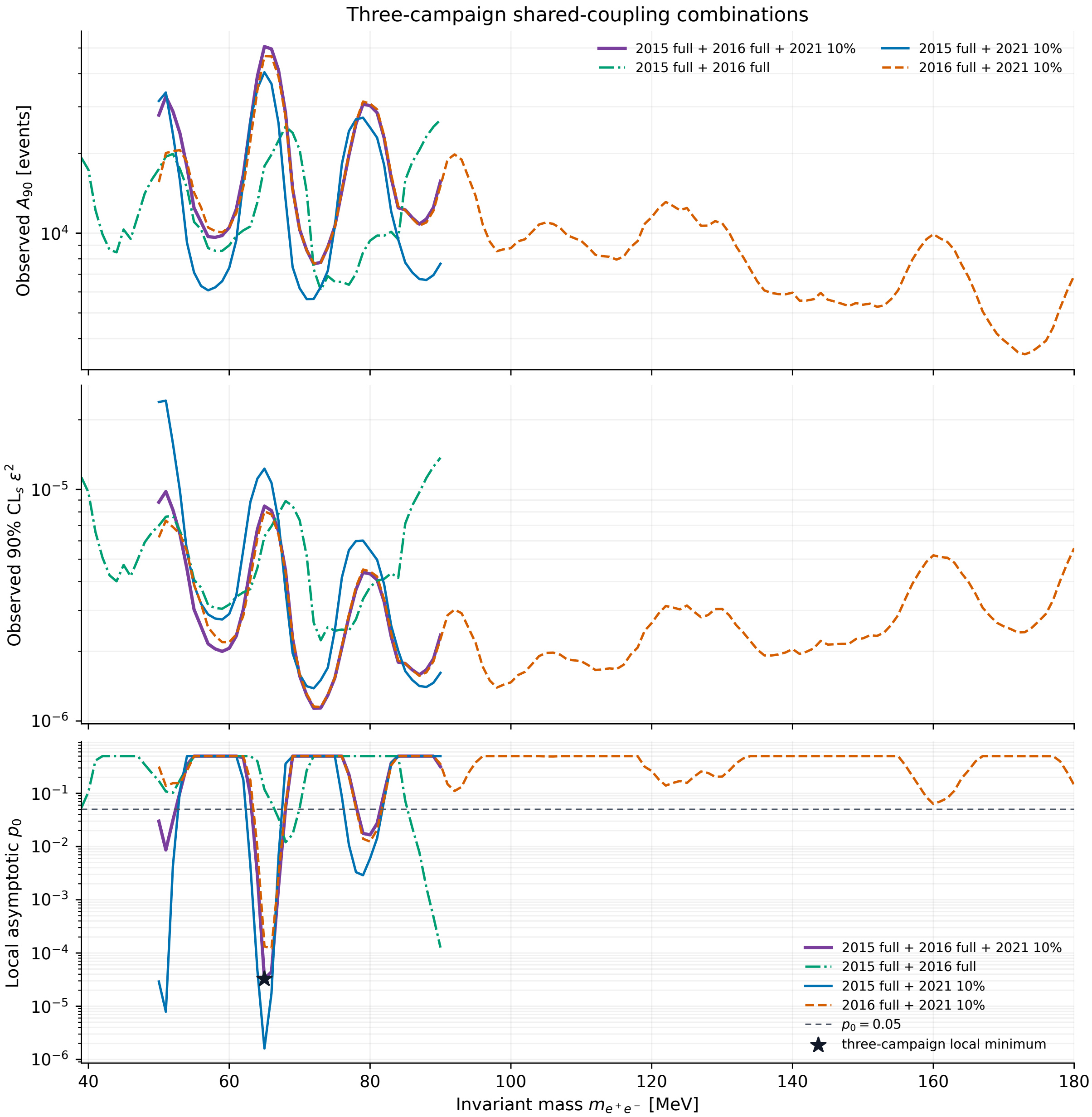


Figure 31: Shared-ε^2 results for the three pairwise combinations and the common three-campaign overlap. For visual alignment, the top coordinate multiplies the shared-ε^2 limit by the sum of the active campaigns' signal normalizations. The remaining panels show the observed 90% asymptotic $\mathrm{CL_s}$ upper limit on ε^2, and the local asymptotic p_0. Each combination uses the three-campaign configuration, including the 40–300 MeV GP support for the 2021 10% contribution. No expected-limit bands are shown.

tests one fixed mass. The series has not been calibrated with scan-wide background-only pseudoexperiments, and no look-elsewhere correction is assigned here. Consequently, the plotted values are local diagnostics rather than global significances. They also should not be read as coverage tests for the upper limits.

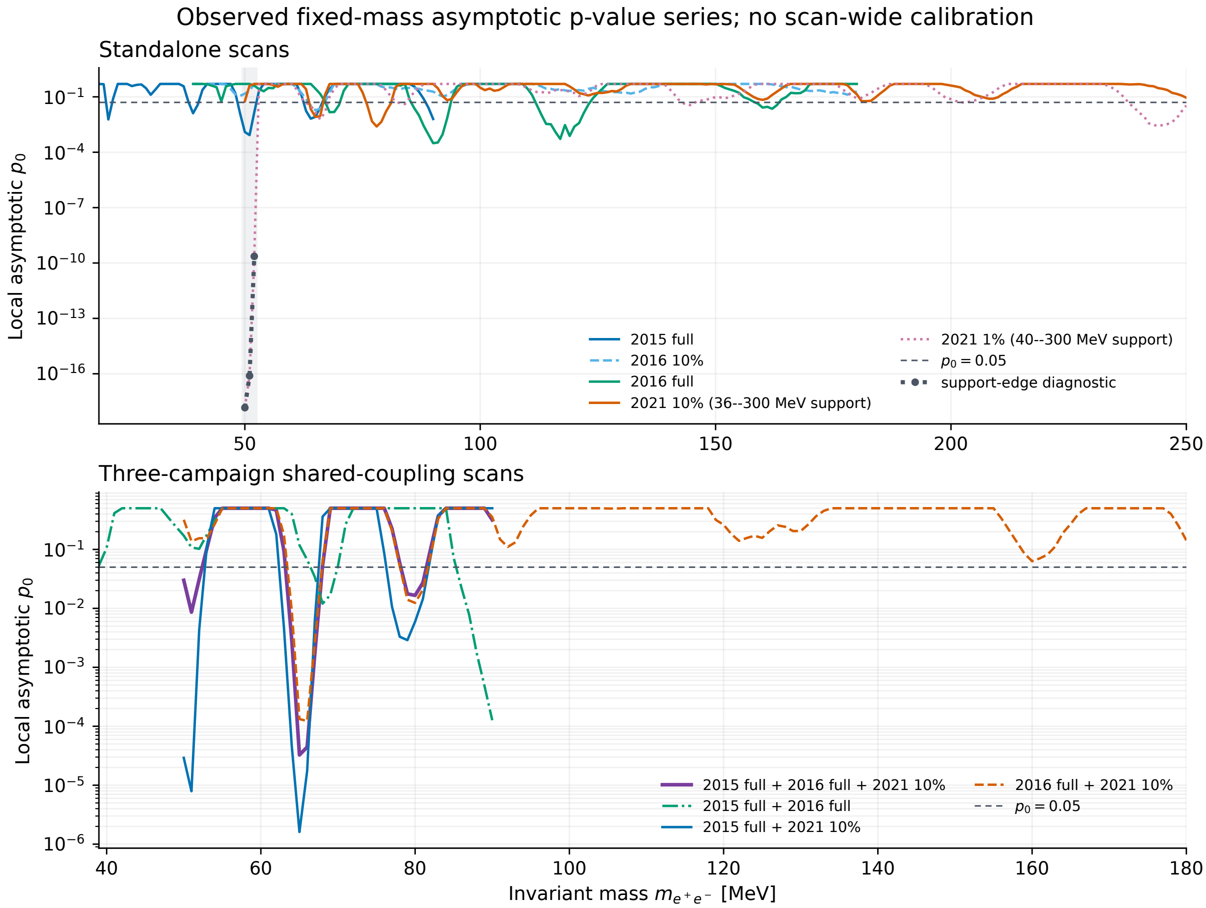


Figure 32: Local asymptotic background-only p_0 as a function of mass for the standalone scans and the three-campaign shared-coupling combinations. The support-sensitive 50–52 MeV segment of the 2021 1% scan is marked as a boundary diagnostic. These are fixed-mass values: no scan-wide calibration or global significance is reported.

6.4 The largest all-three local fluctuation

Within the all-three overlap, the smallest local asymptotic value occurs at 65 MeV: $p_0 = 3.259183 \times 10^{-5}$, corresponding to $Z = 3.99321$ under the one-sided local asymptotic mapping. Figure 33 shows the underlying extraction separately in the three campaigns. Presenting the data, GP background, and fitted signal contribution together makes it possible to see how a single shared coupling is supported—or constrained—by statistically independent spectra.

Combining campaigns increases the statistical information available at a common mass, but the interpretation remains tied to the stated configuration of each curve. A localized excess would require a scan-wide background calibration before it could support a global claim.

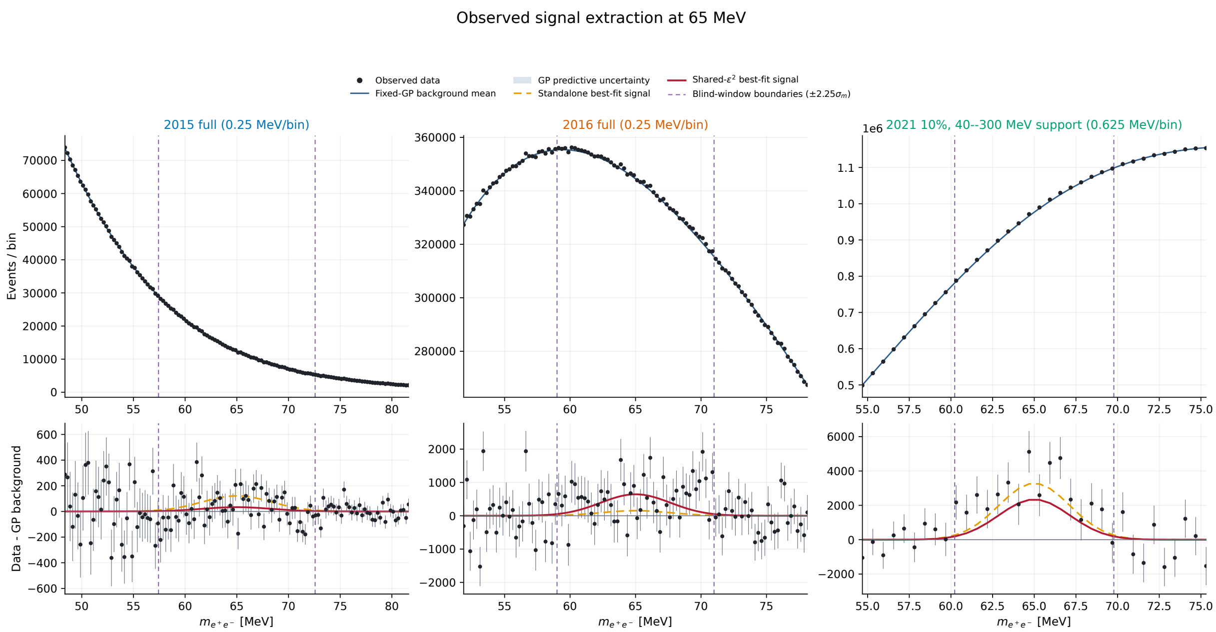


Figure 33: Extraction at the 65 MeV mass hypothesis, the largest local upward fluctuation in the all-three overlap. The three panels show the full 2015, full 2016, and 2021 10% spectra with their GP background descriptions and the signal contributions associated with standalone and shared-ε^2 fits. The quoted p_0 and Z are local asymptotic quantities. This display is a candidate diagnostic, not a scan-calibrated discovery claim.

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